- ome groups of minerals, use in the body, food stuffs conta them and their deficiency disease will be discussed in tabl below:

Minerals	Use in the body	Food stuffs containing them	Deficiency disease
Calcium	• Building bones and teeth • Blood clotting • Activation of certain enzymes	• Milk, fish, cheese	• Weak (fragile) bones and teeth
Phosphorus	• Building of bones and teeth • Making of ATP, DNA and RNA	• Milk, egg, nuts, cheese, meat	• Poor growth of bone and teeth. • Retarded growth
Iron	• Making of hemoglobin • Hemoglobin is the red pigmented substance in RBC's to transport oxygen	• Liver, legumes, meat, egg	• **Anemia**- reduced number of hemoglobin in RBC's which results in general weakness
Iodine	• Used to make a hormone called thyroxin	• Iodized tables salt, vegetables, sea foods	• **Goiter** – enlargement of thyroid gland which results in swelling of the neck
Sulphur	• Constituent of skin, nail, hair.	• Legumes, fish, meat, eggs	

5. **Vitamins**

Vitamins are ***micronutrients***

- Vitamins ar maintenanc
- Lack of vit **disease.**
- Vitamins c
  - ✓ ***Water***
  - ✓ ***Fat sol***
- Some type contain ar below.

Types of vitamin
Vitamin- A (Retinol)
Vitamin-B_1 (Thiamin)
Vitamin-B_2 (Riboflavin)
Vitamin-B_3 (Niacin)

5. Vitamins

Vitamins are other groups of nutrients which are classified as *micronutrients*.

- Vitamins are important for regulation of our body's chemistry, and maintenance of healthy body.
- Lack of vitamins in the diet causes different types of **deficiency disease**.
- Vitamins can be classified as:
  - ✓ ***Water soluble Vitamins*** → are vitamin – C and Vitamin – B groups.
  - ✓ ***Fat soluble vitamins*** → are vitamins A, D, E and K.
- Some types of vitamins, their use in the body, food stuffs they contain and their deficiency diseases will be discussed in table 3.2 below.

Types of vitamin	Use in the body	Food stuffs containing them	Deficiency disease
Vitamin- A (Retinol)	• To make some cell of **retina** in the eye	• Carrots, dairy products, liver, spinach, egg, milk	• ***Night blindness*** is un ability to see in dark or dim light
Vitamin-B_1 (Thiamin)	• Used during cellular respiration	• Cereals, kidney, liver, yeast, milk	• ***Beriberi*** – loss of appetite and nerve inflammation. • Paralysis, mental illness
Vitamin-B_2 (Riboflavin)	• Formation of coenzymes	• Milk, liver, yeast extract, cheese, eggs and green vegetables.	• Sore mouth • Slow growth • Eye disease
Vitamin-B_3 (Niacin)	• Components of coenzymes NAD, NADP.	• Milk, meat, fresh vegetables, whole meal bread, liver, yeast extract	• ***Pellagra*** – characterized by diarrhea, loss of weight, mental disorder

Vitamin-C (Ascorbic acid)	• Sticks together cells lining surfaces of parts such as mouth	• Citrus fruits, green leafy vegetables, tomatoes	• ***Scurvy*** – bleeding of gums. • Failure of wound to heal • Anemia, heart failure
Vitamin-D (Calciferol)	• Growth and strength of bones	• Fish, liver, milk, dairy foods and exposure to sun	• ***Rickets*** – weak bones on parts like knees and legs.
Vitamin-E (Tocopherol)	• Affects muscles and reproductive system	• Green vegetables, cereal oils, milk, egg, liver, seeds	• Sterility in rats
Vitamin-K (phyloquinione)	• Used in blood clotting by synthesizing prothrombin	• Spinach, cabbage,	• Prolonged blood clotting time

Food test

Food test is a laboratory technique that is used to check the presence of nutrients in a given food sample. These can be expressed as:

- **Positive result** – shows the presence of nutrient to be checked in a food sample.
- **Negative result** – shows absence of the nutrient to be checked in a given food sample.

Reagents are chemicals which are used to carry out the food tests.

Table 3.3 summar
result a

Type of nutrient
Starch
Monosaccharide
Disaccharides
Proteins
Fats and oils
Vitamin – C

6. Water

Life on our plane
a constituent of l
Biological signif

- It's a un
substance
dissolve i
reactions
- It has coo
sweating,
regulation
- Involved
hormones
blood.
- Hydrolysi
digestion.
- Removal
- Urine, sw
For

Table 3.3 summarizes some groups of nutrients, reagents used, positive result and negative result.

Type of nutrient	Reagent used	Positive result	Negative result
Starch	Iodine	Blue-black	Brown/Amber
Monosaccharide	Fehling's/ Benedict's	Orange/red	Blue
Disaccharides	Fehling's/ Benedicts	Orange/red	Blue
Proteins	Buriet reagent	Purple to pink	Blue
Fats and oils	Ethanol reagent	Milky suspension or translucent	No emulsion or translucent spot
Vitamin – C	DCPIP (Dichloro phenol (endophenol)	Blue back to colorless	Blue

6. Water

Life on our planet is impossible without water. It is mainly important as a constituent of living cells and habitat for most organisms.
Biological significance of water are:

- It's a universal solvent. It's an excellent solvent for polar substances. Thus most of the substances in our body can easily dissolve in water, accordingly water is a medium for metabolic reactions of the body
- It has cooling effect on our body through evaporation, such as sweating, panting, lubrication which plays role in temperature regulation.
- Involved in transport of substances in the body such as food, hormones, and waste products in the form of solution in our blood.
- Hydrolysis reaction – it hydrolyzes large food molecules during digestion.
- Removal of waste materials from the body in the form of faces. Urine, sweat.
- For Osmoregulation – maintaining water balance of the body for osmotic stability of the body.

Rauphage

Rauphage is a bulk of undigested food materials. Human digesti system cannot digest cellulose which is a component of plant cell wa Thus most of the vegetables (green papers) remain undigested ar called **fiber** or **rauphage**.

- The colon which is part of large intestine absorbs water found i rauphage which is used to facilitate digestive process and avoid **constipation** during removal of feaces.

> ***Constipation*** – is faces that stays too long in the large intestine causes most of its water and becomes compact, hard and difficult to evacuate from the body.

Nutrition

Nutrition is the long process that involves **obtaining** and **utilizing** food. It's a broad process that involves the following steps.

- ***Ingestion*** – taking food in to the mouth
- ***Digestion*** – changing large, complex and insoluble food particles in to their smaller, complex and insoluble food particles.
- ***Absorption*** – is the process in which digested food particles are absorbed in to the blood stream.
- ***Assimilation*** – is utilization of food by our cells.

Based on organisms' mode of nutrition, nutrition can be classified as:

- ✓ **Autotrophic nutrition** – is a kind of nutrition in which organisms synthesize their own food or energy.
- ✓ **Heterotrophic nutrition** – is a nutrition in which organisms cannot make their own food or energy source.

Photo autotroph
- Use sun light t own food.
- This includes g and photosy

$6CO_2+6H_2O \xrightarrow{sunl\ chlor} C_6H_{12}O_6+6O_2$

Heterotrophs can b
- ✓ **Holozoic** Eg. Huma
  - **Sapro** substan **Exampl**
  - **Parasi** organis

Balanced diet

Balanced diet is a f of nutrients approp conditions of the b

Malnutrition

Malnutrition is any can be:

i. ***Under nut*** the body. these come
   - ***Kwashi***
   - discolor
   - ***Marasm***
   - ***Night bl***

ii. **Over nutrit** The typi

Autotrophs	
Photo autotrophs	Chemo autotrophs
• Use sun light to synthesize their own food. • This includes green plants, algae and photosynthetic bacteria. $6CO_2+6H_2O \xrightarrow[chlorophyl]{sunlight} C_6H_{12}O_6+6O_2$	• Use simple chemical reaction to get energy • These include chemosynthetic bacteria. *Example*: Nitrifying bacteria $2NO_2+O_2 \xrightarrow{Nitrifying\ bacteria} 2NO_3+Energy$

Heterotrophs can be grouped in to three: 0

- ✓ **Holozoic nutrition** – involves feeding solid food substances Eg. Humans, Lions.
  - **Saprophytic Nutrition** – Involves feeding organic food substances obtained from dead organisms.
    Examples are decomposers
  - **Parasitic nutrition** – parasites live in and on the surface of organisms (host) by harming them.

Balanced diet

Balanced diet is a food that contains the right type and precise amount of nutrients appropriate for healthy body. It's good for healthy conditions of the body.

Malnutrition

Malnutrition is any kind of abnormality which is related to feeding. It can be:

i. ***Under nutrition*** – if the food eaten is too little than needed by the body. It causes several types of deficiency diseases. Among these come of the mare:
   - ***Kwashiorkor*** – is lack of protein in the diet which may cause discolored hair, thin legs and blotted belly in children.
   - ***Marasmus***- is caused due to lack of carbohydrate.
   - ***Night blindness***- is caused due to the lack of vitamin A.

ii. **Over nutrition**: Eating too much food than needed by the body. The typical example for this is obesity.

- ***Obesity*** is over weightedness that is caused due to excessive consumption of food

Malnutrition can also be characterized by eating wrong types of food only to satisfy our needs,

Illustrative Examples

1. A shortage of vitamin D in the human diet is likely to result in: (EGSECE 1993).

   A. Beriberi C. scurvy
   B. Night blindness D. rickets

   Explanation: **Rickets** – is a deficiency of Vitamin D, which results in poor development of bone in body parts like knee and leg.

   Answer: D

2. Which of the following substances would give a positive test with Benedicts solution? (EGSEC 1993)

   A. Glucose B. Starch C. Fat D. Protein

   Explanation:
   When simple sugars and double sugars react with feelhings/ Benedicts reagent they form orange/ red precipitate. Thus glucose is a simple sugar.

   Answer: A

3. Which of the following is correct about an autotroph? (EGSEC 1993)

   A. Makes its own food
   B. Feeds on dead animal and plant material
   C. Feeds on plants only
   D. Feed on animals only

   Explanation: Autotrophic organisms are organisms that can make their own food. This includes both photosynthetic and chemosynthetic autotrophs.

   Answer: A

4. Which of the following foods has the highest energy content per gram of edible food material? (ESLCE 1994 E.C)

   A. Tomato C. Teff
   B. Groundnut D. Potato

   Explanation:

Food item	Energy contained in KJ/g
Tomato	0.9
Groundnut	24.0
Teff	14.2
Potato	16.9

Answer

5. Loss of appetite and the disease called beriberi are caused by the deficiency of: (EGSECE 1996)
   A. Thiamin
   B. Niacin
   C. Riboflavin
   D. Pyridoxin

Explanation:
Deficiency of vitamin – B_1 (Thiamin) results in loss of appetite and nerve inflammation which may lead to paralysis and mental illness. This condition is called **beriberi.**

Answer: A

6. Which of the following portions of a food is usable and digestible?
   A. Rauphage
   B. Bulk
   C. Fiber
   D. Nutrient

Explanation:
Raughage or fiber is undigested food material.
Nutrients are substances that are usable and digestible in our body. These include carbohydrate, protein, lipid, vitamins and mineral ions.

Answer: D

7. Which of the following deficiency diseases is caused by the lack of vitamin B in our diet? (EGSECE 1997)
   A. Scurvy
   B. Pellagra
   C. rockets
   D. Marasmus

Explanation: Pellagra as a deficiency disease of vitamin B_2 which is characterized by diarrhea, loss of weight and mental disorder.

Answer: B

8. What is the correct order of types of teeth, from front to back? (EGSECE 1997)
   A. canines, incisors, molars, premolars
   B. canines, incisors, premolars, molars
   C. Inscisors, premolars, canines, molars
   D. Incisors, canines, premolars, molars.

Explanation:
Incisor – The front chisel shaped teeth used for cutting lamp of food
Canine – sharp and pointed teeth which in used for biting and tearing flesh.
Premolar and molar – are used for grinding and chewing food.
Molar – is the back teeth.

Answer: D

9. In a food test if an unknown food decolorizes DCPIP, the food contains:
   A. Vitamin C
   B. Protein
   C. Starch
   D. Mineral

   Explanation: If a food that contains Vitamin C, reacts with **DCPIP** (dichlorophenol indophenol) it becomes decolorized and forms a color less solution.

   Answer:

10. Poor bone development is caused by the deficiency of
    A. Retinal
    B. Riboflavin
    C. Tocopherol
    D. Calciferol

    Explanation: Lack of **vitamin – D** (calciferol) causes Rickets which is characterized by weak bone development.

    Answer:

11. If sugar weights of the following substances are oxidized, the largest amount of energy is released by:
    A. Fats B. Proteins C. Starch D. sugars

    Explanation:
    Fat and carbohydrates are energy yielding groups on nutrients. Fats contain highest amount of energy than any other groups of nutrients. Fat contain twice as much energy than carbohydrates

    Answer:

12. The sugar that is found in milk is: (EGSECE 1999 E.C)
    A. Maltose
    B. Sucrose
    C. glucose
    D. lactose

    Explanation:
    ✓ **Lactose** is a sugar that is found in the milk.
    ✓ **Sucrose** is called cane sugar and found in the table sugar
    ✓ **Maltose** is a sugar which is found in germinating cereals like barley

13. If a sample of unknown food changes the color of iodine solution to blue black, it contains:
    A. Protein
    B. Fat
    C. starch
    D. vitamin

    Answer: D

    Explanation:
    When iodine solution is added to a food sample that contains starch, it turns blue black, which shows a positive result for the presence of starch.

    Answer: C

14. Thin legs, bloated belly and discolored hair in children are the signs of the deficiency of:
   A. Vitamin B
   B. Carbohydrate
   C. Vitamin A
   D. protein

Explanation:
Deficiency of protein in the diet causes fluid imbalance, in the body, which is characterized by thin legs, bloated belly and discolored hair in children.

Answer: D

15. Which of the following nutrients is correctly matched with its function? (EGSECE 2001 E.C.)
   A. Proteins – form enzymes
   B. Carbohydrates – regulate metabolism
   C. Vitamins – provide energy
   D. Lipids – repairing damaged fissures

Explanation:
The correct match of the nutrients listed and their function is:
- ✓ **Protein** – form enzymes and repairing damaged tissues
- ✓ **Carbohydrate** – provide energy for the body
- ✓ **Vitamins** – help to regulate metabolic process of the body.
- ✓ **Lipids** – for storage of energy in the body, etc.

Answer: A

16. Which of the following food items has the highest energy content per gram?
   A. Beef
   B. Orange
   C. Wheat
   D. Peanut

Explanation:

Food item	Energy contained in KJ/g
Beef	6.7
Orange	1.8
Wheat	14.2
Peanut	24.0

Answer: D

Item number **17** is based on the following experiment of food test
1. Drop a sample of food on a piece of paper.
2. Dry the paper and expanse it to a light source. The paper became transparent.

17. The sample of food is:
   A. Vitamin
   B. Protein
   C. lipid
   D. Carbohydrate

Explanation:
Test for fat and oil involves:

- ✓ **Oil – spot test** → when a sample of food on a paper fo translucent spot.
- ✓ **Emulsion test** → when a food sample containing fat a oil mixed with water they give milk suspension.

Answe

18. Why is water important to life? It
    A. Exists in three states
    B. Is a medium for metabolic reactions.
    C. Provides body building materials
    D. Is found in large amounts.

Explanation:
Water is important to life in various ways:

- It's a medium for metabolic reactions of the body.
- Its involved in transport of substances
- In removal of was materials from the body
- It hydrolyzes food in digestive activity, etc.

Answer:

Item number 19 is based on the table below. The table shows four children and the type of food they are being fed for a longtime.

Child	Type of food
1	Injera, fish
2	Bread, chicken
3	Rice, bread
4	Rice, milk

19. Which of the children would likely suffer from kwashiorkor?
    Child:
    A. 3 B. 1 C. 4 D. 2
    Explanation: Kwashiorkor is a deficiency disease of protein child 1 gets protein from fish, child 2 from chicken child 4 from milk However, child 3 do not get and food item containing protein because rice and bread are carbohydrates. **Answer: C**
20. Which of the following nutrients form adipose tissue under the skin?
    A. Lipids B. Carbohydrates C. Vitamins D. Proteins

Explana
skin. It's
the skin
21. Which o
    A. O
    ar
    B. C
    C. F
    D. F
Explan
Fat →
Oils →

22. Some p
    body. V
    A.
    B.
Explan
Obesity
special

23. Which
    Organi
    A.
    B.
    C.
    D.
Expla
can sy
24. sunlig
    Whicl
    of cel
    A.
    B.
Expla
becau

Explanation: Adipose tissue is found in the inner layer of the skin. It's made up of lipid or fat and helps to insulate layers under the skin and regulation of body temperature. **Answer: A**

21. Which of the following statements is correct about fats and oils?
    A. Oils are made of amino acids while fats are made of fatty and glyceron.
    B. Oils are solid and fats are liquid at room temperature
    C. Fats are solid and oils are liquids are at room temperature.
    D. Fats dissolve in alcohol while oils dissolve in water

Explanation:
Fat → are solid at room temperature.
Oils → are liquid at room temperature.

Answer: C

22. Some people eat too much and end up with fatty and inflated body. What do you call this condition?
    A. Marasmus
    B. Pellagra
    C. obesity
    D. kwashiorkor

Explanation:
Obesity is overweightednes which results when the food intake specially fat is larger than that of the food output.

Answer: C

23. Which of the following organisms has an autotrophic nutrition? Organisms that: (EGSECE 2003E.C)
    A. Lives on plant bodies
    B. Feeds on small organism
    C. Grows on decayed matter
    D. Uses light to prepare food

Explanation: Autotrophic nutrition involves those organisms that can synthesize their own food or energy. Some outhotrophs use sunlight of make their own food. **Answer: D**

24. Which of the following nutrients serve as structural components of cell membrane?
    A. Vitamins
    B. Carbohydrates
    C. Minerals
    D. Lipids

Explanation: cell membrane is called a **lipo-protein** layer. This is because its made up of protein and lipid bilayer.

Answer: D

25. When we test for monosaccharide with blue feeling's solution a orange precipitate is formed in the test tube. What is the reason o the formation of this orange precipitate? Because the:
A. Monosaccharide is changed to disaccharide
B. Copper in the feheling's solution is oxidized
C. Monosaccharide is reduced in the reaction
D. Copper in the feheling's solution is reduced to copper oxide.

Explanation: simple sugars such as glucose reduce the copper in Fehling'/ Benedict's solution to copper oxide and they themselves are oxidized in the reaction. Therefore, simple sugars or monosaccharide's are called **reducing sugars.**

Answer: D

26. Which of the following is incorrect about macronutrients?
A. Include minerals and vitamins
B. Iron is an essential macronutrient
C. They are needed in small amount
D. Include carbohydrates, proteins and fats.

Explanation: Nutrients can be grouped as macronutrients and micronutrients.
- ✓ ***Micronutrients*** are needed by the body in small amount. This includes vitamins and minerals.
- ✓ ***Macronutrients*** are needed in large amount. This includes protein, carbohydrates and fat.

Answer: D

27. Suppose you are asked to determine Buriet test for a certain food. Which of the following is correct about your test? If the food has.
A. Sugar, orange red color will develop.
B. Protein, orange red color will develop
C. Sugar, purple color will develop
D. Protein, purple color will develop

Explanation: Buriet test is a food test for protein by using buriet reagent (sodium Hydroxide solution and copper sulfate solution) which is mixed and boiled. It finally gives purple color which a positive result for protein tests.

Answer: D

28. List out the major steps of nutrition. .

Explanation:

Nutrition is the process of obtaining and utilizing food. Thus, the major steps of nutrition involves:

Ingestion → Digestion → Absorption → Assimilation

29. Distinguish between autotrophic nutrition and heterotrophic nutrition

Explanation:

- ✓ ***Autotrophic nutrition*** is the process by which organisms synthesize their own food. This includes the major processes such as **photosynthesis** and **chemosynthesis**.
- ✓ ***Heterotrophic nutrition*** is the types of nutrition organisms that cannot synthesize their own food Feed on other organisms

30. Distinguish between macronutrients and micronutrients.

Explanation:

Macronutrients are nutrients that are needed by the body in a large amount. These include **protein, carbohydrate** and **lipid (Fat).**

Micronutrients are nutrients which are required by the body in a very small amount, these are **minerals** and **vitamins**.

31. Why is water so important to life?

Explanation: Water is important to all life form in various ways. These include:

- ➢ It is a medium for metabolic reactions in the body.
- ➢ It dissolves most of the substances in the body.
- ➢ It plays a role in **hydrolysis reaction** which helps to break large sized substances in to smaller pieces, etc.

32. Describe the meaning of malnutrition.

Explanation: **Malnutrition** is any abnormality which is related to the feeding styles. This involves:

- Under nutrition
- Over nutrition
- Eating wrong type of food which does not contain the balanced diet

33. List out at least four properties of water

Explanation: Among various properties of water some of them are:

- It is a universal solvent.
- It has high latent heat of vaporization
- It is transparent
- It has high surface tension, etc.

34. Differentiate between
Saturated and unsaturated fatty acids

Explanation:

Saturated fatty acids	Unsaturated fatty acids
- Contain single bonds between carbon atoms	- Contain double bonds between the carbon atoms.
- Solid at room temperature	- They are liquid at room temperature
- Contain high level of cholesterol	- Contain less amount of cholesterol

Post Test 1

***Part I:* Choose the best answer from the given alternatives.**

1. Which one of the following is a simple sugar?
   A. Lactose C. Starch
   B. Maltose D. Glucose
2. The longest chain of glucose is contained in:
   A. Maltose C. Lactose
   B. Starch D. Sucrose
3. The condensation reaction between two glucose molecules form:
   A. Lactose C. Sucrose
   B. Maltose D. Fructose
4. The correct sequence of steps in nutrition is:
   A. Ingestion → Digestion → Absorption → Assimilation
   B. Digestion → Assimilation → Absorption → Ingestion
   C. Ingestion → Absorption → Assimilation → Digestion
   D. Digestion → Absorption → Ingestion → Absorption
5. The building blocks of protein are:
   A. Fatty acids C. Amino acid
   B. Glycerol D. Glucose
6. Which of the following are micronutrients?
   A. Carbohydrate C. Minerals
   B. Protein D. Fat
7. Nutrients that play role in building and repairing the body are:
   A. Carbohydrates C. Lipids
   B. Proteins D. Vitamins
8. Buriet reagent is used in testing:
   A. Vitamin C. Protein
   B. Mineral D. Carbohydrate
9. Food stuffs that get decolorized up on addition of DCPIP are:
   A. Vitamin A C. Vitamin C
   B. Vitamin B D. Vitamin D

10. A.

Part II:

1. Vita
2. Vita
3. Vita
4. Vita
5. Vita
6. Vita
7. Calc
8. Prot
9. Iron
10. Iodi

3.2 Th

Digestion

particles a
be used up
The proce
ii.
Physical d
smaller pi
✓ th
✓ th
✓ F

The phys
digestive
ii) Ch
Chemical
simpler ar
by enzym
Chemicall
up.

10. Foods such as butter, fatty meat and peanut largely contain:
A. Fat B. Protein C. Carbohydrate D. Water

Part II: **Match the following groups of nutrients with their deficiency diseases.**

	Column A		Column B
1.	Vitamin A	A.	Rickets
2.	Vitamin B1	B.	Goiter
3.	Vitamin B2	C.	Beriberi
4.	Vitamin B3	D.	Kwashiorkor
5.	Vitamin D	E.	Anemia
6.	Vitamin C	F.	Scurvy
7.	Calcium	G.	Pellagra
8.	Protein	H.	Fragile bone
9.	Iron	I.	Night blindness
10.	Iodine	J.	Sore throat

3.2 The Digestive System

Digestion – is the process in which large, complex and insoluble food particles are changed to their smaller, simpler and soluble form that can be used up by our cells.

The process of digestion can be:

ii. **Physical/ Mechanical digestion**

Physical digestion involves breaking down of large food particles in to smaller pieces by

- ✓ the grinding and chewing actions of the teeth,
- ✓ the rolling action of the tongue and
- ✓ Peristaltic movements of the food along the gut.

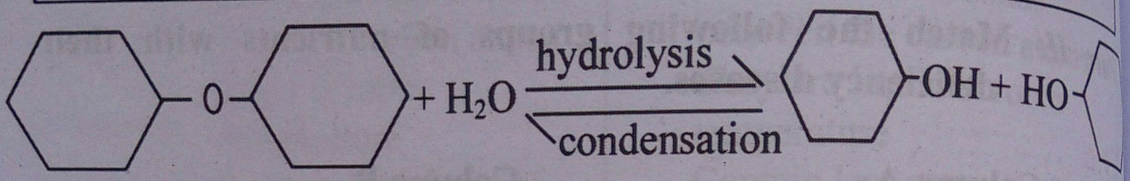
The physical digestion provides much more surface area for the digestive enzymes.

ii) Chemical digestion

Chemical digestion involves converting food particles in to smaller, simpler and soluble forms by the hydrolysis reaction which is mediated by enzymes.

Chemically digested food can be easily absorbed in to the body and used up.

Hydrolysis reaction is a reaction which involves the splitting of large molecules in to smaller pieces (monomers) up on the addition of water.



Enzyme

Enzymes are complex proteins that are produced by the body to bring about changes and speed up (catalyze) the rate of chemical reaction.

- They are important to catalyze various types of reactions that take place inside the body of an organism. Thus, they play a huge role in digestive process.

Therefore, since enzyme play huge roles in facilitating (catalyzing) the chemical reactions of the body, they are called **biological catalysts.**

i) **Extracellular enzymes** are enzymes that work outside the cell.
Examples: Digestive enzymes

ii) **Intracellular enzymes** are enzymes are that work inside the cell.
Example: Respiratory enzymes

Some properties of enzymes

Some of the properties of enzyme include:

- Enzymes are proteins. They are made up of complex proteins,
- Enzymes are biological catalysts. This is because they catalyze chemical reactions inside the living tissue.
- Enzymes are specific. Every reaction of the body is regulated by its own type of enzyme.
- Enzymes can be re-used. This is because enzymes do not take part in the reaction or change in the reaction.
- Enzymes are sensitive to **temperature** and **PH**.
- Enzymes work actively at optimum body temperature. At extremely high temperature enzymes get denatured and at lower temperature than the optimum temperature, they get denature.

- Some
neutra
protea

The Worki

Gut is a hallo
which include
Mouth —
→ *large*
The whole dig
The processes
- **Inges**
(body
- **Diges**
insol
- **Absc**
in to
- **Eges**
rema

Digestion

Mouth is the
the lips.
The food en
which it's br
process is kn
The cavity i
buccal cavit
Both the pl
Physical dig
the teeth, a
digestion in
salivary am

- Some enzymes work best at acidic pH, while other work at neutral pH or basic (alkaline pH. For example, pepsin is a protease enzyme that works best at acidic pH.

The Working of the gut

Gut is a hallow tubular structure which extends from mouth to anus which includes the following pathway.

Mouth ⟶ esophagus (gullet) ⟶ stomach ⟶ small intestine ⟶ large intestine ⟶ Anus

The whole digestive activity starts and ends in the gut.

The processes that occur while, the food moves in the gut are:

- **Ingestion** – the process by which food is take in to the mouth (body)
- **Digestion** – the process of breaking down large, complex and insoluble food in to its smaller, simpler and soluble form.
- **Absorption** – the process by which digested food is absorbed in to the blood stream.
- **Egestion/ defecation** – the process of removing undigested remains of food in the form of feaces.

Digestion in the mouth

Mouth is the anterior opening of the alimentary canal and is bound by the lips.

The food enters the body through the mouth is bittern by the teeth in which it's broken down in to smaller pieces by its chewing action. This process is known as **Mastication.**

The cavity inside the mouth that consists of teeth and tongue is called **buccal cavity.**

Both the physical and chemical digestion starts in the mouth. The Physical digestion is carried out by the grinding and chewing action of the teeth, and the rolling action of the tongue. While, the chemical digestion in the mouth is carried out by the action of an enzyme called **salivary amylase (ptyalin).**

Teeth

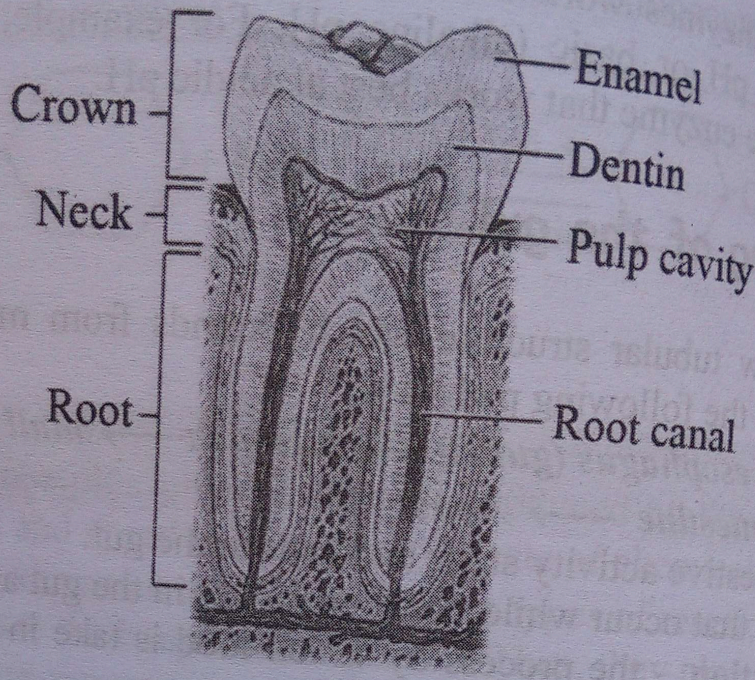


Fig. 3:1 structure of human teeth

The longitudinal section of the teeth has the following parts:

- ✓ **Crown** – the exposed portion of the teeth above the gum
- ✓ **Root** – the part which is embedded in the jaw bone.
- ✓ **Neck** – the narrow region between the root and the crown.
- ✓ **Cement** – the structure that firmly attaches the root into the gum.

The transverse (cross – sectional) area of the teeth consists of three layers. These are:

- ✓ **Enamel** – the structure that covers the crown and neck of a tooth. It's the hardest structure in the body
- ✓ **Dentine** – the hard substance which is found next to enamel. However, it is not as hard as enamel.
- ✓ **Pulp cavity** – the inner sensitive layer of the teeth that consist of blood vessels and nerves. It is sensitive to heat, cold pain, etc.
  - It is also called *'the living tissue'* of the teeth.

Types of teeth

In mammals including human being, there are four types of teeth. These are:

i) **Incisor** – the front chisel shaped a tooth that is used to cut the hard lamp of food.

ii) **Canine** – is the sharp and pointed teeth which is used for biting and tearing flesh.

iii) **Premolar** – is used for grinding and chewing food.

iv) **Molar** – is the back teeth which is used for grinding and chewing food.

Teeth can be easily exposed to bacterial infection. This is because bacteria can easily get the sugary food which is dissolved with saliva in the mouth. They form a thin film known as **plaque**. These bacteria produce acid that can dissolve the enamel and dentine; finally they reach the inner sensitive layer of the teeth which is called pulp cavity which results in tooth ache or pain. It also makes the teeth to become discolored. This condition in which the teeth become painful and discolored is called **tooth decay.**

Periodontal disease is the bacterial infection of the gums which causes bleeding and loss of teeth.

The tooth should be kept clean and healthy by:

- Regular brushing of teeth and gums twice a day.
- Avoiding sweet, sugary food.
- Regular dental checkups

Finally the food will be digested:

- ✓Physically by the chewing and grinding actions of the teeth and the rolling action of the tongue.
- ✓Chemically by the action of an enzyme called **salivary amylase** or **ptyalin** which is produced by the salivary gland.

$$\text{Starch} + H_2O \xrightarrow{\text{ptyalin}} \text{Maltose}$$

Then, a mass of food known as **bolus** passes in to the esophagus.

Movement of the food along the esophagus

Esophagus is a narrow tube which is about 25 centimeter long that carries the swallowed food in to the stomach. Its walls are muscular. Thus, the contraction and relaxation of the esophagus wall brings further mechanical breakdown of the food.

When the tongue pushes the food in to the pharynx, nerves stimulate the swallowing reflex. The food then enters the esophagus or food tube, and is moved to the stomach by rhythmic muscular contractions called **peristalsis.**

Swallowing is a reflex action that takes place when the food reaches the back of the throat.

The entrance of food in to the wind pipe (trachea) is prevented by epiglottis.

The food passes from the esophagus to the stomach through a ring muscle called **sphincter**.

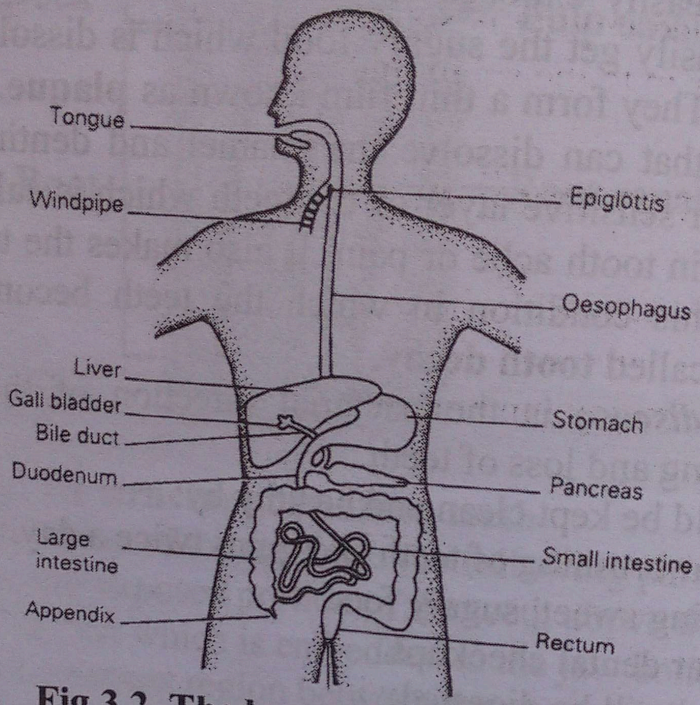


Fig 3.2. The human digestive system

Digestion in the stomach

The lower esophageal sphincter relaxes when the food reaches it, allowing the food to enter the **stomach.**

The stomach is a muscular sac (bag) in which the food is collected and partial digested by protease enzymes.

Note: **Protease enzymes** are proteins that digest proteins.

Digestion in the stomach is assisted by a solution called **gastric juice** which is secreted by gastric wall (gland).

Gastric juice is a solution that consists of **hydrochloric acid (HCl), mucus** and **protease enzymes** (pepsin and Rein).

- ✓ **HCl** is used to kill germs; as well as, to make the stomach suitably acidic for the action of enzymes called **pepsin** and **renin**.
- ✓ **Protease enzymes** are enzymes that are used to catalyze digestion of protein. These are **pepsin** and renin.

$$\text{Protein} + H_2O \xrightarrow{pepsin} \text{peptones}$$

✓ **Mucus** protects muscles walls of the stomach from being digested by protease enzymes and attack by the acid.

After few hours of digestion; thick, grey, soup like mixture of partially digested food called **chyme** is squeezed out of the stomach through the **pyloric sphincter** in to the small intestine.

Digestion in the small intestine

Small intestine is a place where most of the digestion of food takes place. It has three parts; namely;

✓ Duodenum, jejnum **and** ileum.

i) Duodenum

Duodenum is the upper part of small intestine where the digestive process is facilitated by the secretions from the accessory organs like the **liver** and **pancreas**.

The liver is an organ that works with the duodenum by secreting a salty solution called **bile** which is stored in the gall bladder. Bile is used to emulsify fat in to fat droplets.

$$\textbf{Fat} \xrightarrow{\textbf{Bile}} \textbf{Fat droplets}$$

➢ The pancreas is a tongue shaped organ (gland) which lies beneath the stomach and surrounded on one side by the curve of duodenum. It secretes a solution known as **pancreatic juice** which consists of enzymes called **trypsin, pancreatic amylase** and **lipase**.

The action of enzymes in the duodenum is discussed as follows:

$$\text{Starch} \xrightarrow{\text{pancreatic amylase}} \text{maltose}$$

$$\text{Peptones} \xrightarrow{\text{Trypsin}} \text{peptides}$$

$$\text{Fat droplets} \xrightarrow{\text{lipase}} \text{fatty acids} + \text{Glycerol}$$

The roles of bile during the digestion in the duodenum are:

➢ It emulsifies fat in to fat droplets.

➢ It neutralizes the acidity of the chime.

Remember: The liver and pancreas are not the parts of the gut (alimentary canal). However, they help the digestive process in the duodenum. The liver produces the bile and the pancreas secretes a number of digestive enzymes.

ii) Jejnum is the part of the small intestine that joins the duodenum with the ileum.

iii) Ileum

Ileum is the lower, the longest and the narrower part of the small intestine. It has both digestive and absorptive role.
The digestion in the ileum is assisted by a solution called **intestinal juice** which is secreted by the **intestinal** glands. The Intestinal juice consists of enzymes that bring about the end products of digestion. These are; ***Sucrase, Maltase, lactase, erepsin*** and ***lipase.***

Maltose $\xrightarrow{\text{maltase}}$ Glucose + Glucose

Sucrose $\xrightarrow{\text{Sucrase}}$ Glucose + Fructose

Lactose $\xrightarrow{\text{Lactase}}$ Glucose + Galactose

Fat droplet $\xrightarrow{\text{Lipase}}$ Amino acids

Peptides $\xrightarrow{\text{erepsin}}$ Fatty acids + glycerol

Table 3: Summary of enzymatic actions in the human digestive system

Part of the gut	Glands	Secretions	pH	Enzymes	Reaction
Mouth	Salivary gland	Saliva	7	Salivary amylase (ptyalin)	Starch + H_2O ***amylase*** maltose
Stomach	Gastric gland	Gastric juice	2	Pepsin Renin	Protein pepsin peptones cougulates milk protein.
Duodenum	Pancreas	Pancreatic	7	Amylase Typsin Lipase	Starch + H_2O ***amylase*** maltose peptones Trypsin peptides Fat *Lipase* Fatty acid + glycerol
Ileum	Intestinal gland	Intestinal juice	8.5	Maltase Lactase Surcease Lipase Erepsin	Maltose ***maltase*** glucose +glucose Lactose ***lactase*** glucose + galactose Sucrose ***sucrase*** glucose + fructose Fat ***Lipase*** Fatty acid + glycerol Peptides ***erepsin*** aminoacid

Absorption

Absorption is th
from the walls o
The lower porti
digested food
surrounded by
and the lymph

35. Enzyme
A.
B.
Explan
regulat
made

36. In ma
stoma
A.
B.
Expl
of en

ends
ere

37. Bile
dro
A.
B.
Ex
liv
wh

38.

Absorption of digested food

Absorption is the process by which the end products of digestion pass from the walls of the small intestine and enter the blood stream.
The lower portion of the small intestine is well adapted for absorption of digested food since it has millions of finger like projections that are surrounded by network of blood capillaries called **villi** (singular **villus**) and the lymph vessel called **lacteal.**

Illustrative Examples

35. Enzymes are made up of (GSCE 1993 E.C)
    A. Fats C. Proteins
    B. Carbohydrates D. Mineral salts
    Explanation: Enzymes are biological catalysts which are used to regulate or speed up the chemical reactions in the body. They are made up of complex proteins. **Answer: C**

36. In mammals, which of the following starts being digested in the stomach and has its digestion finished in the small intestine?
    A. Starch C. sugar
    B. Fat D. protein
    Explanation: Protein digestion starts in the stomach by the action of enzyme called **pepsin.** Protein + H_2O $\xrightarrow{pepsin}$ peptones and ends in the small intestine by the action of **erepsin** peptides $\xrightarrow{erepsin}$ amino acids. **Answer: D**

37. Bile salts are used to break up fats into small, easily digested droplets. This process is called
    A. Absorption C. digestion
    B. Emulsification D. excretion
    Explanation: Bile is a salty substance which is secreted by the liver. Its stored in gall bladder: It works in digestive activity which takes place in the duodenum. It helps for:
    - Neutralizing the acidity of chyme
    - Emulsification of fat.

    Fat $\xrightarrow{bile}$ Fat droplets **Answer: B**

38. There are many kinds of enzymes in a single cell because
    A. Enzymes form enzyme – substrate complex.
    B. Enzymes are made up of amino acids.
    C. Enzymes are very specific
    D. Enzymes can be used over and over again.

Explanation: Enzymes have variety of properties. One of the properties states that, enzymes are specific; this statement explains that there are many kinds of enzymes in cells that are adapted to catalyze their own specific type of reaction.

39. Which of the following enzymes act on proteins in the stomach? **Answer:**
A. Renin C. lactose
B. Peptidase D. Lipase

Explanation:
Renin is an enzyme that works for protein digestion in children. It helps to solidify milk protein in children.

40. Which of the following organic compounds undergo partial chemical digestion in human mouth? (EGSECE 1996 E.C) **Answer: A**
A. Carbohydrates C. proteins
B. Fat D. minerals

Explanation: Chemical digestion of carbohydrates starts in the mouth by the action of an enzyme called **salivary amylase** or **ptylin**. $\text{Starch} + H_2O \xrightarrow{\text{amylase}} \text{maltose}$ **Answer: A**

41. Which of the following portions of a food is usable and digestible?
A. Rouphage B. Bulk C. Fiber D. Nutrient

Explanation: Nutrients are usable and digestible portions of the food that are absorbed in to the blood stream in the small intestine. While, rauphage/ fiber or bulk are undigested remains of the food.

42. Which of the following are produced as a result of the mechanical digestion of a piece of meat? **Answer: D**
A. Amino acids C. small meat
B. Peptides D. peptons

Explanation: *Mechanical digestion* – is a kind of digestion which involves break down of large food particles into smaller pieces without bringing any kind of chemical change.

43. Which of the following protein digesting enzymes is found in the stomach? **Answer: C**
A. Pepsin C. Rickets
B. Erepsin D. Marasmus

Explanation: Pepsin and rennin are protease enzymes which are found in the stomach **Answer: A**

44. Which of the following digestive process is affected by the gallbladder is surgically removed?
A. Digestion of lipids
B. Digestion of carbohydrates
C. Digestion of proteins
D. Digestion of Vitamins

Explanation: Gallbladder is a storage site of bile which is used for emulsification of fat **Answer: A**

45. Saliva contains an enzyme that begins the digestion of
A. Protein
B. Disaccharide
C. starch
D. peptides

Explanation: saliva consists of an enzyme ptyalin that changes starch into maltose. **Answer: C**

46. Which statement describes all enzymes? (EGSECE 1998 E.C)
A. They control the transport of materials
B. They provide energy for chemical reactions
C. They affect the rate of chemical reactions
D. They absorb oxygen from the environment

Explanation: Enzymes affect the rate of chemical reactions by speeding it. **Answer: C**

47. In a food test if an unknown food decolorizes DEPIP the food contains.
A. Vitamin C
B. Protein
C. starch
D. mineral

Explanation: DCPIP colorizes the food sample if it contains vitamin-C **Answer: C**

48. Which of the following is the main function of the large intestine?
A. Absorption of sugars
B. Absorption of fatty acids
C. Absorption of amino acids
D. Absorption of water

Explanation: large intestine has part which is known as colon that is used to absorb water from fiber or rauphage. **Answer: D**

49. Which of the following is not an enzyme although it plays role in digestion?
A. Lipase
B. Bile
C. Renin
D. Amylase

Explanation: Bile is a salt produced by the liver to emulsify fa... to fat droplets.

Answer: B

50. "Each kind of enzyme catalyzes one kind of reaction." T... implies that enzymes are: (EGSECE 1999 E.C)
   A. Sensitive to PH
   B. Specific
   C. proteins
   D. proteins

Explanation: Enzymes are specific. This means, every reaction ... the body is catalyzed by its own specific type of enzyme.

Answer: ...

51. Which of the following is part of the small intestine?
   A. Ileum
   B. Rectum
   C. colon
   D. Anus

Explanation: small intestine has three major parts. These ar... **duodenum, Jejnum** and **ileum.** Ileum has both digestive an... absorption role.

Answer: A

52. Which of the following end product of digestion pass in to lacteals during absorption of food?
   A. Vitamin
   B. Fatty acids
   C. glucose
   D. amino acids

Explanation: Absorption of glucose, amino acids and vitamins takes place in **villi** which are finger like projections in the **ileum**. **Fatty acids** and **glycerol** are too large to be absorbed by villi, thus they are absorbed in to **lacteals**.

Answer: B

53. Under which of the following conditions does the enzyme ptyline works best?
   A. PH7 and 30^0c
   B. PH7 and 37^0c
   C. PH2 and 300c
   D. PH5 and 37^0c

Explanation: Enzymes are sensitive to PH and temperature, thus ptylin an enzyme that works best at nutrealpH7 and 37^0c, which is human body temperature.

Answer: B

54. Which of the following food items is correctly matched with its digestive enzyme and product of digestion?
   A. Peptones $\xrightarrow{\text{trypsin}}$ peptides
   B. Proteins $\xrightarrow{\text{lipase}}$ amino acid
   C. Fats $\xrightarrow{\text{bile droplets}}$ fatty acids and glycerol
   D. Starch $\xrightarrow{\text{pepsin}}$ Maltose

Explanation: The correct match of food items indicated and products is:

- Fat droplets $\xrightarrow{\text{lipase}}$ fatty acid and glycerol
- Starch + H_2O $\xrightarrow{\text{ptylin}}$ Maltose
- Peptides $\xrightarrow{\text{erepsin}}$ amino acids

Answer: A

55. Digestive system of humans, the pyloric sphincter is located between the:
    A. Gallbladder and intestine
    B. Nasal and mouth cavity
    C. Stomach and intestine
    D. Esophagus and stomach

Explanation: pyloric sphincter is found in between the stomach and intestine, because its located at the lower end of the stomach.

Answer: B

Items number 49 and 50 are based on the table below which shows different reactions that take place in the alimentary canal of humans.

No	Reaction	Product
1	Lipids $\xrightarrow{Bile}$	Fat droplets
2	Protein + H_2O $\xrightarrow{Pepsin}$	Peptones
3	Starch + H_2O $\xrightarrow{Amylase}$	Maltose
4	Peptide + H_2O $\xrightarrow{?}$	Amino acids

56. Where does the reaction indicated by number 1 take place? In the
    A. Ileum
    B. Mouth
    C. duodenum
    D. stomach

Explanation: Lipids are changed in to fat droplets by the action of bile in the duodenum. The whole chemical digestive process of lipid ends here, in the duodenum.

Answer: C

57. What is the enzyme that causes the digestion indicated by number 4?
    A. Ptyalin
    B. Pepsin
    C. Typsin
    D. Erepsin

Explanation: Erepsin is a protease enzyme which converts peptide in to amino acid. This phase of protein digestion takes place in the ileum.

Answer: D

58. Distinguish between mechanical digestion and chemical digestion.

Explanation: **Mechanical (physical) digestion** is the process of cutting large food particles in to smaller pieces.

Chemical digestion is the process of changing in soluble and complex food particles in to smaller pieces by enzymatic action.

59. List out the layers of human teeth from outer to inner one.

Explanation: Enamel → Dentine → pulp cavity

60. Explain the meaning of peristalsis.

Explanation: **Peristalsis** – is the movement of the food along the gut it's muscular contraction and relaxation.

61. List out the contents of the following secretions of the human gut.

- ✓ Gastric juice
- ✓ Intestinal juice
- ✓ Pancreatic juice

Explanation:

Solution	Secreted by	Secretions
Gastric juice	Gastric glands	HCl, mucus, pepsin, rennin
Pancreatic juice	Pancreas	Lipase, trypsin, amylase
Intestinal juice	Intestinal gland	Lipase, sucrase, maltose, lactase, erepsin

62. What is the importance of rauphage in the human digestive system?

Explanation: **Rauphage** is undigested remains of food that passes down from small intestine to the large intestine. It is sued to facilitate the digestive process and avoid constipation since it contains excess water.

63. Describe the structures which are not parts of the gut but play role in assisting the human digestive system.

Explanation: **Liver** and **pancreas** are not parts of the gut or alimentary canal, even though they play role in digestion. The liver secretes the **bile**, while the **pancreas** produces pancreatic juice which contains enzymes.

64. Trace the pathway of the human gut.

Explanation: Mouth → esophagus → stomach → small intestine → large intestine → Anus

65. List out at least three examples for:

- ➢ Protease enzymes
- ➢ Carbohydrate enzymes

Explanation:

- ***Protease enzymes*** are protein digesting enzymes.
  Examples: pepsin, rennin, trypsin, erepsin.
- ***Carbohydrase enzymes*** are enzymes that digest carbohydrates.
  Examples: Amylase, surcease, lactase, maltase

66. Describe the functions of the following parts of the large intestine.
✓ Colon
✓ Rectum

Explanation: **Colon** reabsorbs water from the rauphage which helps to:

- ➢ Facilitate the digestive process.
- ➢ Avoid constipation.

Post Test 2

Part I: Choose the best answer from the given alternatives.

1. Which one of the following is a protease enzyme?
   A. Lipase C. Sucrase
   B. Erepsin D. Amylase
2. The digestive process that takes place in the mouth is:
   A. Starch $\xrightarrow{\text{amylase}}$ Maltose
   B. Maltose $\xrightarrow{\text{maltase}}$ glucose
   C. Protein $\xrightarrow{\text{pepsin}}$ Peptone
   D. Fat $\xrightarrow{\text{lipid}}$ Fatty acid & glycerol
3. Which one of the following is a solution of gastric juice?
   A. Lipase C. Pepsin
   B. Trypsin D. amylase
4. All of the following are properties of enzymes except:
   A. are specific C. can be reused
   B. are carbohydrates D. are catalysts
5. Enzymes such as sucrase, lipase, maltose and erepsin are secreted by:
   A. Pancreas C. Gastric wall
   B. Intestinal wall D. salivary gland

6. An enzyme that coverts starch in to maltose is:
   A. Pepsin C. Lactose
   B. Maltase D. Amylase
7. The part of human digestive system where the process of digestion ends and absorption takes place is:
   A. Duodenum C. Jejunum
   B. Ileum D. Esophagus
8. The hardest protective layer of the teeth is:
   A. Dentine C. Pulp cavity
   B. Enamel D. Crown
9. A salty substance in the duodenum that is used emulsify fat is:
   A. Chyme B. HCl C. Bile D. Feaces
10. The role of liver in human digestive system is:
   A. Secretion of bile C. secretion of saliva
   B. Producing HCl D. emulsification of fat

3.3 The respiratory system

The respiratory system is vital in order to get oxygen into the body to oxidize food and release energy for the proper functioning of the body. Accordingly all organisms need to exchange gas, even though they have variation in complexity and structures they use. Some single called organisms use the simple diffusion for exchange of gas.

The Human Respiratory System

The human respiratory system involves two important processes, these are **inhalation** and **exhalation**.

Inhalation is the process of taking air into the lung.

Exhalation is the process of expelling or removing air from out of the lung.

The human respiratory system consists of the following structures.

- ***Mouth*** is an organ through which the air enters in to the respiratory structures.
- ***Nose (Nasal cavity)*** – is additional structure through which the air enters in to the respiratory tract. The nose has large surface area, good blood supply, lots of hairs and lining that secretes mucus. Thus, breathing through the nose is good for the health of our respiratory system than the mouth.

- ***Pharynx*** is the structure that serves as a common passage for both the food and air.
- ***Trachea*** is the wind pipe which is made from C – shaped rings of cartilage. This helps to avoid collapsing of the wind pipe. The opening of the trachea is called **glottis**. Thus, the entrance of food and dust particles in to the glottis is prevented by the cartilaginous flap called **epiglottis.** The epiglottis closes when someone swallows the food; this prevents the entrance of the food in to the trachea.
  The upper part of the trachea is called **larynx**. The larynx is called the voice box, this is because it is where the sound is produced.
- ***Bronchi (Singular, Bronchus)*** are the two branches that + arise from the trachea, one leading to each lung.
- ***Bronchioles*** are small tubes branching from each bronchus in the lung.
- ***Alveoli*** are millions of tiny air sacks inside the lung, which are sued for exchange of gases between the lung and the blood capillaries. They are called the functional units of the lung.

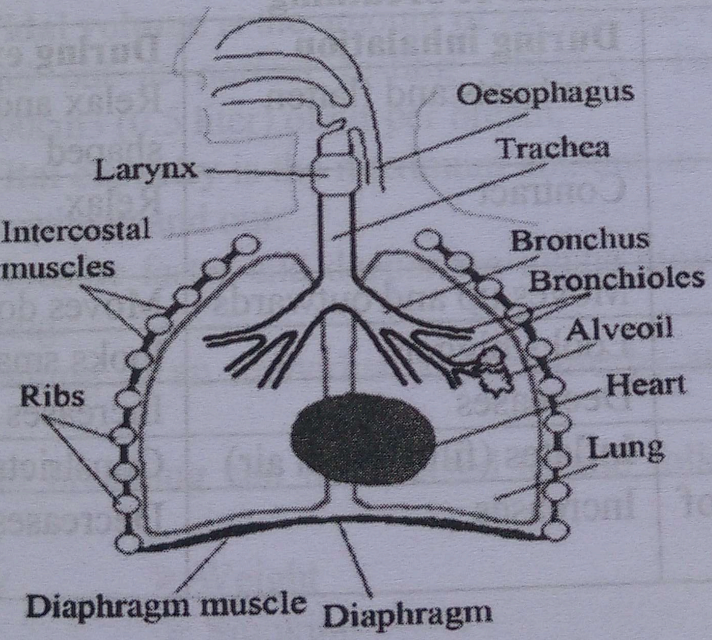


Fig. 3.3: Human breathing system

The lung

The lung is the major breathing organ of most animals including humans. It is surrounded by a membranous structure called pleura membrane.

The lung is spongy and elastic organ which is protected by ribs, **sternum, vertebral column, diaphragm** and **intercostals muscles**.

- **Intercostal muscles** are muscles that are found between the ribs.
- **Diaphragm** is the muscular sheet extending a cross the body cavity between the thorax and abdomen.

The Mechanism of Breathing

The process of breathing involves the process of inhalation and exhalation. During this process the breathing system is assisted by the movement of various structures of the body. These movements are discussed below in table 4.

Table 4: The mechanism of breathing

	During inhalation	During exhalation
Diaphragm	Contracts and flatten	Relax and becomes dome shaped
Intercostals muscle	Contract	Relax
Ribcage	Moves up and outwards	Moves down and inwards
Chest cavity	Looks bigger	looks smaller
Pressure	Decreases	Increases
Lung	Inflates (filled with air)	Constricts (deflates)
The volume of thorax	Increases	Decreases

The Comparison of Inhaled and Exhaled air

Our atmosphere is composed of the mixture of various gases. These gases are mainly **nitrogen**, **oxygen** and **carbondioxide**

Thus, the composition of inhaled and exhaled air will be discussed in the table below.

Table 5: Composition of inhaled and exhaled air

Gas	Composition in inhaled air in percentage	Composition of exhaled air in percentage
Nitrogen (N_2)	78%	78%
Oxygen (O_2)	21%	16%
Carbondioxide (CO_2)	0.04%	4%
Water vapor	less	high
Heat	less	more

Factors that affect breathing

The total capacity of an inflated lung is about 5 liters. However, we do not efficiently use the maximum capacity of our lung. Our breathing is determined by the rate of breathing and depth of breathing.

i. **Rate of Breathing** is the number of breath per unit time.
The normal rate of breathing in adult human being is 12-14 breathes per minute.

ii. **Depth of breathing** is the amount of air per breath.
- ✓ **Tidal volume** is the amount of air that one can breath in and out at normal resting situation. This is about 500cm3 (0.5 liter) of air per breath.
- ✓ **Vital capacity** is the maximum amount air that one can breathe in and out.

Therefore, as we breathe faster and deeper we inhale move amount of oxygen in to our lung.

The depth of breathing of an individual is measure by an instrument called **spirometer.**

The normal rate of breathing can be affected by the following major factors. These are:

- ➢ Anxiety
- ➢ Drugs
- ➢ Altitude
- ➢ Weight
- ➢ Smoking

1. Exercise

During exercise the rate and depth of breathing increases in order to supply more oxygen to release energy for the body.

Exercise makes the muscular activity to increase, accordingly blood supply to our muscles increase to bring more glucose and oxygen to the tissues efficiently.

Regular exercise is important to make a person to have increase rate and depth of breathing. In addition regular exercise has various benefits. These include:

- To get healthy muscular structure and body weight
- To make the muscle strong.
- To have an increased blood supply to the muscles in order to bring more glucose (sugar) and oxygen to the muscles by avoiding muscle fatigue.
- To have an increased size of heart and lung which have efficient blood supply

2. Anxiety

During anxiety the body reacts as it is in danger, thus extra oxygen is needed to release more energy in order to surviving danger. Therefore, the rate and depth of breathing increase.

3. Drugs

Drugs are chemicals that can bring changes in the proper working of the body. Thus they may have stimulating, depressive or hallucinating effects.

Stimulant drugs such as khat, amphetamines and cocaine can increase the rate and depth of breathing

Depressants such as alcohol lower the rate of breathing which makes our cells to be deprived of oxygen supply. This may have serious effects on the body.

4. Altitude

Altitude is the height of the land above sea level. As an altitude increase the air becomes less dense. This means volume of air in a given space becomes lower; this results in declining the amount of oxygen.

At places of higher altitude the level of oxygen becomes lower and lower this makes breathing difficult. Thus, the rate and depth of breathing becomes higher, especially for someone from the place of lower altitude.

People who live at the place of higher altitude do not face the problem which is related to the lack of oxygen, this is because they have a lung which has many more alveoli and more red blood cells to efficiently transport oxygen fastly to the cells in the body.

5. Weight

Over weightiness increases the accumulation of fat on abdominal cavity. This makes it difficult for the diaphragm and other structures around the lung to contact and relax properly. This makes people unable to breathe properly and bring sufficient amount of oxygen in to the lung.

6. Smoking

Smoking is the bad hat can largely affect the respiratory system and other organs to the body. This is because the cigarette smoke consists of around 4000 chemicals that are inhaled in to the lung and can affect health. Some of these include:

i) **Carcinogenic substances** are cancer causing substances that may cause **lung cancer,** and other related cancers of mouth, lip and throat. The most commonly known carcinogenic substances are **arsenic** and **benzeprene**.
ii) **Tar** is the residue of cigarette smoke that accumulates in the lung and cause irritation of nose, throat and lung.
iii) **Nicotine** is the powerful addictive chemical that is found in the cigarette smoke.
iv) **Carbon monoxide** is a very poisonous substance that decreases the oxygen carrying capacity of the blood.

Effects of Smoking on the Health

Smokers' health may get affected in various ways. Some of the health problems that are related with smoking include:

- Cancers of lung, lip and throat can be caused due to carcinogenic substances that are found in the cigarette smoke. About 90% of the lung cancers related to smoking.
- **Chronic obstructive pulmonary diseases (COPD)** results in breaking down of alveoli that reduces surface are for gas exchange in the lung.

- **Bronchitis** is inflammation of bronchi. This causes production of the mucus which leads to the blockage of passage in to the bronchioles. This makes the breat difficult.
- Smoking also affects on the heart and blood vessels wh increases the risk of heart attacks and stroke.

Effects of smoking on individuals, family and society

Smoking may have severe effects individuals, family and the society. Some of its effects include:

- Economic crisis on an individual and family.
- Psychological problems
- Increased risk of respiratory diseases
- Conflict in the family, etc.

Hygiene of breathing

There are mechanisms that can be sued to keep the breathing system in to a healthy state. These are:

- ✓ Good personal hygiene through cleaning teeth, tongue and other respiratory structures.
- ✓ Good ventilation reduces the risk of droplet infection.
- ✓ Avoid smoking cigarette.
- ✓ Covering your mouth during cough and sneezing.
- ✓ Consult a doctor for any problems related to respiratory organs.

Artificial respiration

Breathing activity of a person may fail because of many reasons. These include **drowning, heart attack, accidents, etc.** Thus, when the breathing of an individual fails, we need to give assistance to restore his/her breathing.

The mechanism of restoring breathing to an individual whose breathing is lost is called **artificial respiration**.

Therefore, the
1) Call fo
2) Check uncon
3) Open might
4) Check puttin

66. The pa
A.
B.
C.
D.

67. Air le
A.
B.
C.
D.

Expl

G
N
C
C

68. The
A.
B.

Exp
whe

69. cap
W
(EC
A.
B.
C.
D.

Therefore, the procedures to be followed are:

1) Call for help loudly
2) Check to see if the causality is conscious and if he/she is unconscious call for help.
3) Open the air way by removing obstacles from mouth which might block the air way, if any.
4) Check for breathing once again and check for the causality by putting your head on the nose or mouth if he/she is breathing.

Illustrative Examples

66. The path followed by air as it enters the human body is:
   A. Nose →Trachea → Bronchioles →Bronchus →Alveolus
   B. Nose→Bronchus→Trachea →Bronchioles →Alveolus
   C. Nose →Bronchus → Bronchioles →Trachea →Alveolus
   D. Nose →Trachea → Bronchus → Bronchioles →Alveolus

Answer: D

67. Air leaving the lung differs from air entering the lungs by having
   A. Less oxygen but equal amount of carbondioxide.
   B. More oxygen and less carbondioxide
   C. Less oxygen and more carbondioxide
   D. Less oxygen and less carbondioxide

Explanation: comparison of inhaled and exhaled air

Gas	Composition in inhaled air %	Composition in exhaled air %
N_2	78%	78%
O_2	21%	16%
CO_2	0.04%	4%

Answer: C

68. The most important process in the lungs takes place in the
   A. Pleural cavity
   B. Alveoli
   C. Bronchioles
   D. Bronchi

Explanation: **Alveolus are the smallest air sacks in the lung where exchange of gas takes place between the lung and blood capillaries.**

Answer: B

69. Which of the following happens during inhalation? (EGSECE 1996 E.C)
   A. Rib muscles relax
   B. Size of the chest cavity decreases
   C. Diaphragm contracts
   D. The air pressure inside the lung increases

Explanation:

	During inhalation	During exhalation
Rib muscles	Contracts	Relaxes
Size of chest	Increases	Decreases
Diaphragm	Contracts	Relaxes
Air pressure	Decreases	Increases

Answer:

70. Which of the following diseases of the breathing system is caus by smoking cigarettes?

A. Pneumonia
B. Tuberculosis
C. Influenza
D. Emphysema

Explanation: Emphysema is inflammation of bronchioles whi results in breaking down of alveoli. This is caused due to tox chemicals found in cigarette smoke. **Answer:**

71. What happens to the concentration of gases in the air wi increasing altitude?

A. The percentage of gases will vary
B. The amount of CO_2 will remain the same
C. The amount of oxygen will reduce
D. The composition of air will change

Explanation: As an altitude increases the air becomes less dense. Thus there is less oxygen in a given volume of air at higher altitude. **Answer: C**

72. Which of the following parts of the respiratory tract is made up of rings of cartilages?

A. Trachea
B. Lung
C. Alveoli
D. Bronchiole

Explanation: Trachea is a windpipe which is made up C-shaped rings of cartilage. This helps to keep the tube open for the passage of air. **Answer: A**

73. In human respiratory system, which of the following parts is nearest of the mouth? (EGSECE 1998E.C)

A. Windpipe
B. Larynx
C. Epiglottis
D. Bronchial tubes

Explanation: The pathway of human respiratory system, starting from mouth is indicated as:

Mouth ⟶ pharynx ⟶ epiglottis ⟶ larynx
alveoli ← Bronchioles ← Bronchi ← Trachea ←

Answer: C

74.

75.

76

74. What happens in our thoracic cavity when the diaphragm contracts?

A. The lung expands being filled with air.
B. The size of the cheast cavity decreases
C. The diaphragm gets dome shaped structure
D. The outside air pressure increases as compared to the inner one

Explanation:
During inhalation contraction of the diaphragm and other related structures expands and fills the lung with air.

Answer: A

75. Clean air should **NOT** contain

A. Oxygen
B. Nitrogen
C. Carbondioxide
D. Carbonmonoxide

Explanation:
Carbonmonoxide is a poisonous gas that reduces the oxygen carrying capacity of the blood which may result in death.

Answer: D

76. If you hold cobalt chloride paper about 3cm from your mouth and breath on it for 2 and 3minutes, the paper turns pink. This shows that exhaled air contains.

A. Carbondioxide
B. Water vapor
C. heat
D. Oxygen

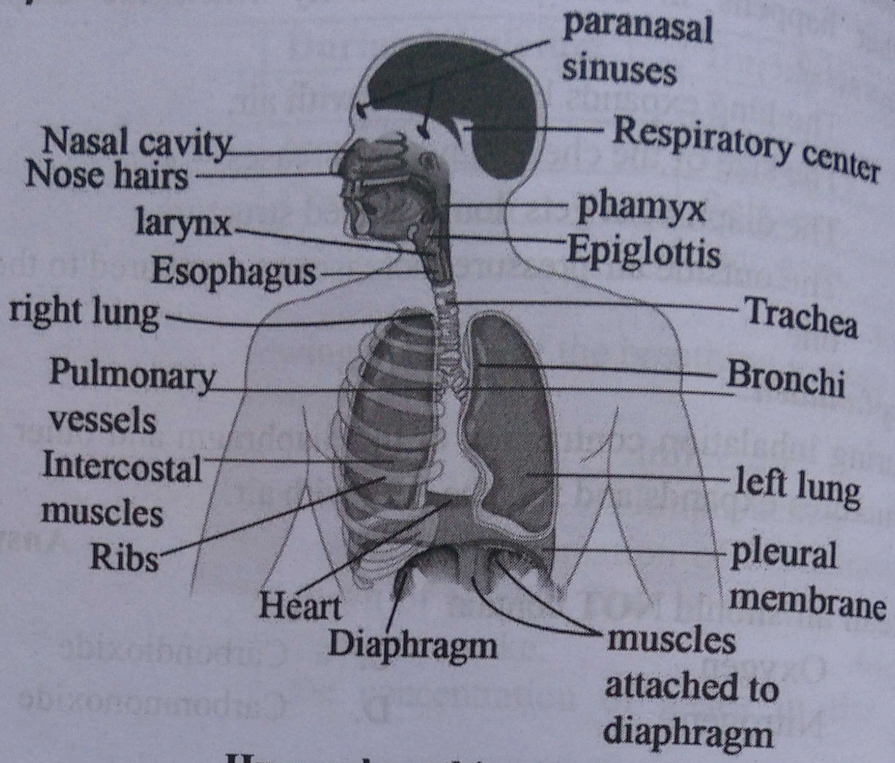
Explanation:
If water is dropped on blue cobalt chloride paper it turns it's color in to pink. Thus exhaled air turns the blue cobalt chloride paper into pink color. This shows the presence of water vapor in exhaled air.

Answer: B

77. In which structure of the lungs does exchange of gases with the blood take place? (EGSECE, 2005 E.C)

A. Capillaries B. Bronchioles C. Alveoli D. Bronchi

Explanation:



Human breathing system

- **Alveoli** – are tiny air sacks in the lung which are surrounded by net work of blood capillaries.
  - They are adapted for exchange of gas between the lung and blood.

78. Which of the following components of cigarette smoke may cause a bleeding in the brain (stoke)? **Answer: C**

A. Tar
B. Arsenic
C. Benzidine
D. Nicotine

Explanation:

Nicotine acts up on brain and nervous system. In smoker's blood it causes narrowing of blood vessels. This may damage the heart and may cause stroke (bleeding in the brain)

79. People living at higher altitude have more red blood cells than people living at lower altitudes. This is because air at higher altitudes contains. **Answer: D**

A. More oxygen in a given volume of air.
B. More carbondioxide in a given volume of air
C. Less oxygen in a given volume of air.
D. Less carbon dioxide in a given volume of air.

Explanation: people living at higher altitude have increased number of red blood cells, this is because the air is less dense at the place of higher altitude. Therefore, in order to supply oxygen

fastly to their body's cells, they have more red blood cells to transport it.

Answer: C

80. What causes muscle fatigue after a very hard physical exercise?
A. Lactic acid
B. Acetic acid
C. carbonic acid
D. oxalic acid

Explanation:
During heavy exercise, human body uses anaerobic respiration to get more energy. Therefore glucose molecules split to form lactic acids and 2ATPs

$$C_6H_{12}O_6 \longrightarrow \underset{\text{lactic acid}}{2C_3H_6O_3} + 2ATP$$

Answer: A

81. Tidal volume of air is the air which is:
A. Inhaled after forced exhalation
B. Exchanged during a normal breath
C. Forcibly exhaled after normal inhalation
D. Remains in the lung after forced exhalation.

Explanation:
Tidal volume is the air which is inhaled and exhaled at normal condition when we are at resting situation.
Vital capacity – is the maximum amount of air that we can inhale and exhale during conditions like heavy exercises.

Answer: B

82. Which of the following happens during breathing air in to the lung?
A. The ribs move down and inwards.
B. The diaphragm becomes dome shaped
C. The volume of chest cavity increases
D. The intercostals muscles relax.

Answer: C

83. In the activity to demonstrate the tar in cigarette smoke, the white cotton wool plugged in the smoking machine changes to brown color as result of tar accumulation. Where does such effect observed in the smokers body? In the
A. Lungs
B. Brain
C. kidneys
D. heart

Explanation:
Even though the rest parts of the body can also be affected by tar in cigarette smoke, lung is the main breathing organ that is largely affected.

84. What happens when the diaphragm gets its dome shape? (EGSECE2002 E.C)

A. Air is withdrawn from the lungs.
B. Air is inhaled in to the lungs.
C. The volume of chest cavity increases
D. The pressure in the lung will decrease

Answer:

Explanation:

	During inhalation	During exhalation
Diaphragm	Flattens	Becomes dome shaped
Air	Inhaled into lung	rushes out of the lung
Pressure in the lung	Decreases	Increases
Volume of chest	Increases	Decreases

85. Which of the following methods would you use to test the presence of carbondioxide in exhaled air? Breathing on:

A. Thermometer
B. Cobalt chloride paper
C. glass surface
D. lime water

Answer: A

Explanation:

When carbondioxide reacts with lime water or bicarbonate indicator solution turns cloudy/yellow, indicating the presence of carbondioxide.

86. The advantage of breathing through the nose is that it (EGSECE 2001E.C)

A. Requires less energy to breath
B. Warms and moistens the air
C. Helps to obtain plenty of air
D. Helps to take air at faster rate

Answer: D

Explanation:

It's better to breathe through nose rather than breathing through the mouth. This is because the nostrils consists of cilia and mucus to prevent entrance of dust particles and also the cilia in the nose moistens and warmth the air.

87. Which of the following breathing structures in located more near to the lung?

A. Larynx
B. Pharynx
C. Glottis
D. Bronchus

Answer: B

Explanation: Nasal cavity → pharynx → Glottis → larynx → bronchus → lung

Answer: D

88. Which of the following actions increases the volume of chest cavity?
   A. Lowering of rib cage
   B. Contraction of diaphragm
   C. Relaxation of diaphragm
   D. Relaxation of the intercostal muscles

Explanation:
During inhalation, the diaphragm contracts and the volume of chest cavity increases.

Answer: B

88. Trace the pathway of the air incoming in to the lung.

Explanation: The pathway of the incoming air into the human lung is:

Nose and mouth →pharynx →Trachea →Bronchi→ Bronchioles →Alveoli

89. List out the major structures that protect our lung.

Explanation: Our lug is one of the delicate organs which can be protected from damage by:

- ✓ Ribs
- ✓ Vertebral column
- ✓ Diaphragm
- ✓ Intercostals muscles

90. Differentiate between:
   - ✓ Rate of breathing and depth of breathing
   - ✓ Tidal volume and vital capacity

Explanation:
Rate of breathing refers to number of breath per unit time; while, depth of breathing is the amount of air per breath.

- ✓ Tidal volume is the amount of air that we breathe at normal situation, while vital capacity in the maximum amount of air that one can inhale and exhale.

91. List out the major factors that affect the normal functioning of the human breathing system.

Explanation:
The human breathing can be affected by factors such as exercise, altitude, smoking, disease, anxiety, etc.

92. List out some of the components of the cigarette smoke that affect our health.

Explanation: Smoking cigarette is obviously known to affect the health of the smoker because it contains dangerous chemicals such as **tar, nicotine, carciriogenic substances** and **carbon monoxide**

93. What are the effects of smoking on the smokers' health?

Explanation: Smoking cigarette may cause the following health problems to the smoker. These include:

- ✓ Cancers of lip, throat, lung and other organs.
- ✓ Chronic obstructive pulmonary disease (COPD)
- ✓ Bronchitis and emphysema, etc.

94. Distinguish between passive smoke and active smoker.

Explanation:

- ✓ **Passive smoker** is a person who sits or lives with the smoker.
- ✓ **Active smoker** is a person who is directly involved in the smoking.

95. What is artificial respiration?

Explanation: **Artificial respiration**: is the method of restoring breathing to an individual whose breathing activity is stopped.

Post Test 3

Part I: Choose the best answer from the given alternatives.

1. The pathway of organs of human breathing system during inhalation is:
   - A. Pharynx→nasal cavity →Glottis→larynx→bronchus → lung
   - B. Nasal cavity →glottis →pharynx→larynx→bronchus → lung
   - C. Nasal cavity →pharynx→Glottis→larynx→bronchus → lung
   - D. Lung →pharynx→Glottis→larynx→bronchus → nasal cavity
2. During exhalation:
   - A. Diaphragm relaxes
   - B. Rib cage moves in and downwards
   - C. Lung inflates
   - D. Chest cavity increases

3. All of the following factors affect the rate of breathing, except:
   A. Altitude C. Light
   B. Exercise D. Drug
4. Exchange of gas between blood capillaries and lung takes place in the:
   A. Glottis B. Trachea C. Alveoli D. Pharynx
5. The maximum amount of air that one can breathe in and out is:
   A. Vital capacity C. Normal breathing
   B. Tidal volume D. Air volume
6. An instrument that is used to measure the depth of breathing is:
   A. Stethoscope C. Sphygmomanometer
   B. Spirometer D. None of the above
7. The powerful addictive chemical found in cigarette smoke is:
   A. Tar C. Arsenic
   B. Nicotine D. Carbon monoxide
8. As altitude increases:
   A. the composition of oxygen increases
   B. the air becomes dense
   C. the composition of oxygen decreases
   D. the air becomes thinner

3.4 Cellular respiration

Respiration – is process in which energy is released from the breakdown of organic substances in the body. Energy that is produced during cellular respiration is supplied to our cells in the form of ATP

What is ATP?

ATP – is an abbreviation which stands for **Adenosine tri-phosphate.**

– Is the direct or immediate source of energy for our cells.

– It's also called 'the cash form of energy'

$$ATP \longrightarrow ADP + Pi + energy$$

ATP is formed by the bond between Adenosine di-phosphate ADP free inorganic phosphate group (p).

Energy that is released from the breakdown of glucose molecules locks itself in a bond that is formed between ADP and phosphate.

When energy is needed by our cells ATP molecule is **hydrolyzed** releases energy stored with in it for our cells

$$ADP + Pi \underset{\text{Energy required}}{\overset{\text{Energy produced}}{\rightleftharpoons}} ATP$$

Types of respiration

Based on the requirement of oxygen, respiration can be classified as:

i. **Aerobic respiration** – is a kind of respiration that requires oxygen.
ii. **Anaerobic respiration** – does not require oxygen to release energy from glucose molecule.

Anaerobic respiration in microorganisms

Alcoholic fermentation is a typical example of anaerobic respiration carried by unicellular fungi called **yeast.**
The reaction for alcoholic fermentation is:

$$C_6H_{12}O_6 \xrightarrow[fermentation]{Yeast} \underset{\text{ethanol}}{2C_2H_5OH} + \underset{\text{carbondioxide}}{2CO_2} + 2ATP$$

In this reaction yeast releases 2ATP from one glucose molecule. Alcoholic fermentation is important for making alcoholic drinks such as, Tej, Tella, Beer.

Anaerobic respiration in humans

Normally humans use aerobic respiration. But during heavy exercise since our body needs energy, much oxygen is required by our cells to carry aerobic respiration.
If someone is stranger to such kinds of heavy exercise, its difficult for him or her to inhale sufficient amount of oxygen needed by our cells to oxidize glucose molecule. Thus glucose molecule starts to break in to two lactic acid molecules and release energy needed by the cells.

$$C_6H_{12}O_6 \longrightarrow \underset{\text{Lactic acid}}{2C_3H_6O_3} + 2ATP$$

Lactic acid produced accumulates on the muscle and causes pain on muscles called –muscle **fatigue** or **scramp.**

Lactic acid is a poisonous chemical, so it should be removed from our muscles, thus oxygen is needed to break it in to carbondioxide and water.

Oxygen dept – the amount of oxygen needed to break down lactic acid.

Lactic acid + oxygen ⟶ carbondioxide + water

Aerobic respiration

Aerobic respiration is efficient form of respiration that provides more energy than anaerobic respiration.

It involves release of energy from oxidation of organic substances.

$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 36/38$ ATP

Comparison of aerobic and anaerobic respiration

Anaerobic respiration	Aerobic respiration
✓ Do not utilize oxygen ✓ Less amount of energy is produced. Only 2ATPs from 1 glucose molecule ✓ It mostly occurs in the cytoplasm of a cell	✓ Utilizes oxygen ✓ Relatively high amount of energy is produced. 36/38 ATPs from one glucose molecule. ✓ It always occurs in mitochondria of a cell.

Illustrative Example

96. When compared with anaerobic respiration, the amount of energy released by aerobic respiration is:

A. More C. less

B. Equal D. unknown

Explanation: During aerobic respiration about 36/38ATps are released from one glucose molecule, while during anaerobic respiration only 2ATPs are released from one glucose molecule.

Answer: A

97. The tubes that split from the trachea and lead in to left and right lungs are known as: (EGSECE, 2006 E.C)

A. Trachioles C. bronchi

B. Alveoli D. bronchioles

Explanation:

- **Trachea** – is the main wind pipe that carries air down in to the bronchi.

- **Bronchi** – are the two branches that split from the trach and leading air in to each lung.
- **Bronchioles** – are smaller branches that arise fro bronchi.
- **Alveoli** – are tiny air sacks where exchange of gas betwe the lung and blood capillaries takes place.

Answer:

98. The direct energy source of cellular respiration is:
A. Lactic acid C. glycogen
B. ATP D. glucose

Explanation: Adenosine triphosphate (ATP) is the only form o energy that can be directly used by our cells, accordingly its also called 'the cash form of energy. **Answer:** B

99. What causes muscle fatigue after a very hard physical exercise? (EGSECE 1999 E.C)
A. Lactic acid C. carbonic acid
B. Acetic acid D. oxalic acid

Explanation: During heavy exercises if we cannot inhale sufficient amount of oxygen to get more energy, our body starts to carry out anaerobic respiration in which glucose molecule splits to form lactic acid which is poisonous and causes muscle fatigue. **Answer:** A

100. Which of the following equations is correct during formation of chemical energy from food?
A. ATP + phosphate + Energy $\longrightarrow$ ADP
B. ATP + phosphate + ADP $\longrightarrow$ energy
C. ADP + phosphate $\longrightarrow$ ATP + energy
D. ADP + phosphate + energy $\longrightarrow$ ATP

Explanation: ATP is formed from ADP (Adenisine diphosphate), free phosphate group and energy that is released from the breakdown of glucose molecule. **Answer: D**

101. The anaerobic stage of cellular respiration in aerobic organisms is known as:
A. Kreb's cycle C. electron transport
B. Glycolysis D. citric acid cycle

Explanation:
Glycolysis – is the first stage of cellular respiration in which glucose molecule splits to form 3 carbon molecule called pyravate. This process occurs in the cytoplasm and its called the **anaerobic stage**. It produces 2 Net ATPs. While the rest stages indicated are aerobic stage and they occur in the mitochondria.

Answer: B

102. The fermentation of glucose by yeast normally Yields
A. Lactic acid, CO_2 and 2ATP
B. Alcohol, CO_2 and 36 ATP.
C. Alcohol, CO_2 and 2ATP
D. CO_2, H_2O and 36 ATP

Explanation: Yeast carries anaerobic respiration whenever there is no oxygen available. The process is called **alcoholic fermentation**. The chemical reaction is summarized as:

$$C_6H_{12}O_6 \xrightarrow[fermentation]{Yeast} 2C_2H_5OH + 2CO_2 + 2ATP$$
(alcohol)

Answer: C

103. Which of the following organelles contain the enzymes needed to synthesize ATP in the presence of oxygen?
A. Mitochondria
B. Endoplasmic reticulum
C. Ribosomes
D. Golgibodies

Explanation: Mitochondria is called 'the power house of the cell' because it's the main site of aerobic respiration in our cells.

Answer: B

104. Which of the following mineral elements is important as a component of ADP and ATP?
A. Phosphorus
B. Magnesium
C. potassium
D. chlorine

Explanation:
The phosphates found in ADP and ATP contain phosphorous.

Answer: A

105. The usefulness of fermentation as a means of deriving energy is limited because:
A. It cannot generate enough ATP.
B. It produces too much CO_2.
C. The end products are toxic to the body.
D. It causes more energy than it produces

Explanation:
During fermentation only 2 ATPs can be generated from glucose molecule. This is the least efficient amount of energy that can be produced from one glucose molecule.

106. The advantage of anaerobic respiration is that **Answer: A**
A. It is more efficient than aerobic respiration.
B. Energy can be produced when oxygen is in short supply.
C. Many ATP molecules are produced from each molecule of glucose
D. Pyruxic acid is converted to lactic acid.
Explanation: Aerobic respiration relies on sufficient supply of oxygen; while anaerobic respiration do not require oxygen to release energy.

107. Which condition would most directly result in the production of lactic acid by muscle cells in the human body? **Answer: B**
A. Excess amount of nitrogen in the cells.
B. Insufficient amount of CO_2 in the cells.
C. Insufficient amount of nitrogen reaching the cells.
D. Insufficient amount of oxygen reaching the cells.
Explanation: Lack of sufficient oxygen in to the cells of animals makes them carryout anaerobic respiration which produces lactic acid.

Answer: D

Items number 107 is based on the following table which shows the process of respiration. (EGSECE 2001 E.C)

Reaction	1	2	3	4
Reaction	Glucose	Pyruvic acid	Glucose	Pyruvic acid
Product	Lactic acid	Ethyl alcohol	Carbondioxide	Carbindioxide
Amount of energy produce	2ATP	2ATP	36 ATP	36 ATP

108. The correct process of anaerobic respiration is shown by:
A. 3 B. 4 C. 1 D. 2
Explanation: Reaction number 1 shows anaerobic respiration in animals.
Glucose ⟶ lactic acid + 2ATP

Answer: C

109. Which of th
cellular acti
A. Pyru
B. Gluc

Item 110 is based
different organism

Cell
A
B
C

110. Cell 'C' is
A. Ye
B. Hu
Explanat
Glucose -

Item number 87
$C_6H_{12}O_6$
111. The reac
A. A
B. A
C. A
D. A
Explana
microor
112. Which
A.
B.
C.
D.
Explan
in anim
yeast.
113. A cell
A.
B.

109. Which of the following molecules is the direct energy source for cellular activities?
A. Pyruvic acid
B. Glucose
C. Acetic acid
D. Adenosine triphosphate

Answer: A

Item 110 is based on the table given below which shows respiration in different organisms cells.

Cell	Provided with	Produce
A	Glucose	CO_2,H_2O,38ATP
B	Glucose	Lactic acid, 2ATP
C	Glucose	CO_2, ethyl alcohol, 2ATP

110. Cell 'C' is more likely a:
A. Yeast cell
B. Human cell
C. fern cell
D. bird cell

Explanation: Anaerobic fermentation by yeast is:

$$\text{Glucose} \xrightarrow[\text{fermentation}]{\text{Yeast}} \text{ethyl alcohol} + \text{carbon dioxide} + 2\text{ATP}$$

Answer: A

Item number 87 is based on the following respiratory reaction.

$$C_6H_{12}O_6 \xrightarrow{\text{enzyme}} 2C_2H_5OH + 2CO_2 + 2ATP$$

111. The reaction represent:
A. Anaerobic respiration in microorganisms.
B. Anaerobic respiration in animals
C. Aerobic respiration in plants.
D. Aerobic respiration in animals

Explanation: Yeast is a single celled fungus which is classified as microorganisms.

Answer: A

112. Which of the following are products of anaerobic respiration?
A. Oxygen and water
B. Water and carbon dioxide
C. Ethanol and water
D. Lactic acid and ethanol

Explanation: lactic acid is produced during anaerobic respiration in animals; while ethanol is produced by anaerobic respiration in yeast.

Answer: D

113. A cell with large number of mitochondria is expected to:
A. Make a lot of ATP
B. Produce more protein
C. store more food
D. perform photosynthesis

Explanation:
Mitochondria is the site of cellular respiration where ATPs are released. Thus, a cell with large number of mitochondria is expected to make a lot of ATPs.

Answer:

Item number 113 is based on the following reactions.

Reaction 1: glucose $\longrightarrow$ lactic acid + ATP

Reaction 2: Glucose $\longrightarrow$ Ethanol + CO_2 + ATP

114. Which of the following is NOT true?
   A. Reaction 2 is anaerobic
   B. Reaction 1 is aerobic
   C. Reaction 2 occurs in yeast cells
   D. Reaction 1 occurs in muscle cells.

Explanation:
Reaction number I is anaerobic respiration in animals.

Answer: B

115. All of the following process utilizes ATP except
   A. Anabolic process of the cell
   B. Muscular contractions that affects movement
   C. Catabolic process of the cell.
   D. The diffusion of oxygen from alveoli to blood capillaries.

Explanation:
Diffusion is the movement of molecules from an area of higher concentration of lower concentration. It's a passive transport system that takes place without utilization of energy (ATP).

115. What is ATP?

Answer: D

Explanation:
ATP (Adenosine triphosphate) is a chemical substance that directly supplies energy from our cells

116. Clearly differentiate between aerobic respiration and anaerobic respiration

Explanation:

Aerobic respiration	Anaerobic respiration
✓ Utilizes oxygen ✓ Produces less amount of energy	✓ Do not utilize oxygen. ✓ Produces high amount of energy

117. What causes muscle fatigue a very hard physical exercise?
Explanation:
Muscle fatigue which is caused after a very hard physical exercise is due to accumulation of lactic acid on our muscles.

Post Test 4

Part I: Choose the best answer from the given alternatives.

1. Which one of the following is true about aerobic respiration?
   E. 2 ATPs are produced from one glucose molecule
   F. Takes place in the absence of oxygen
   G. Lactic acid is produced
   H. Takes place in the presence of oxygen
2. A substance that causes muscle scramp after having heavy exercise is:
   A. Acetic acid C. Carbon dioxide
   B. Lactic acid D. Methane
3. The anaerobic respiration in yeast produces:
   A. Glucose B. Lactic acid C. Ethanol D. Oxygen
4. Which of the following reaction shows the correct reaction about ATP?
   A. $ATP + ADP \rightarrow Pi$
   B. $ADP + Pi \rightarrow ATP$
   C. $ATP + Pi \rightarrow ADP$
   D. $Pi + ADP \rightarrow ATP + Pi$
5. The process of alcoholic fermentation is:
   A. Anaerobic respiration by humans
   B. Aerobic respiration by yeast
   C. Aerobic respiration by human
   D. Anaerobic respiration by yeast

3.5 The circulatory system

The transport system is required to supply the needs of the body cells and remove the waste materials they produce.
The blood circulation system is the main transport system in humans. It consists of:

i. Heart
ii. Blood vessels
iii. Blood

The blood Vessels

They are pathways through which transport takes place. There are three main types of blood vessels. These are

- ✓ Artery, Vein, Capillaries

i) Artery

Artery is a blood vessel that is used to carry blood away from the heart. Most arteries carry oxygenated blood except **pulmonary artery** and **umbilical artery**.

- ✓ **Pulmonary artery**- carry deoxygenated blood away from the heart to the lung.
- ✓ **Umbilical artery** – carry deoxygenated blood away from the foetus to placenta.

Arteries:

- ✓ have thick and elastic wall
- ✓ Are largest of artery is **aorta** while the smallest arteries are called **arterioles.**
- ✓ Have no valves.

ii)Vein

Veins return blood back to the heart.
Most veins carry deoxygenated blood, except **pulmonary vein** and **umbilical vein**.

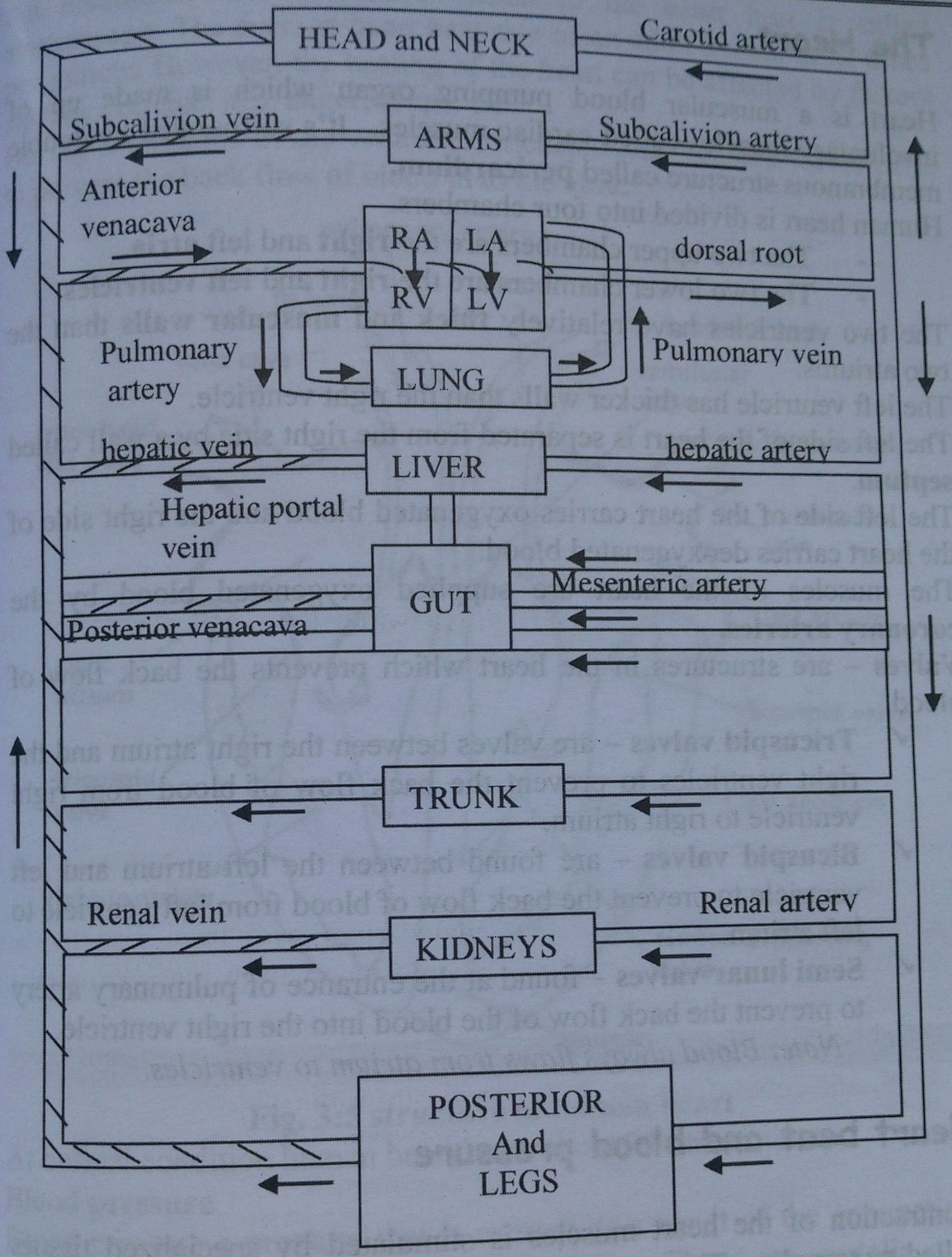
- **Pulmonary vein** – return oxygenated blood back from lung to heart.
- **Umbilical vein** – carries oxygenated blood from placenta back to the developing fetus.

Veins:

- ✓ have relatively thin and less elastic wall.
- ✓ Are largest vein is called **venacava,** while the smallest veins are called **venuoles.**
- ✓ have valves

ii) Capillaries

Capillaries are narrow, thin – walled blood vessels. The capillaries help to connect arteries with veins and take blood to the tissues and cells. They have no valves.



▨ Deoxygenated blood

== oxygenated blood

Fig. 3:4 the main components of human circulatory system

Key

RA = right atrium
LA = left atrium
RV = right ventricle
RL = left ventricle

The Heart

Heart is a muscular blood pumping organ which is made up of involuntary muscles called cardiac muscles. It's surrounded by double membranous structure called **pericardium.**

Human heart is divided into four chambers.

- The two upper chambers are the **right** and left **atria.**
- The two lower chambers are the **right** and **left ventricles.**

The two ventricles have relatively **thick** and **muscular walls** than the two atriums.

The left ventricle has thicker walls than the right ventricle.

The left side of the heart is separated from the right side by a wall called **septum.**

The left side of the heart carries oxygenated blood and the right side of the heart carries deoxygenated blood.

The muscles of the heart are supplied oxygenated blood by the **coronary arteries.**

Valves – are structures in the heart which prevents the back flow of blood.

- ✓ **Tricuspid valves** – are valves between the right atrium and the right ventricles to prevent the back flow of blood from right ventricle to right atrium.
- ✓ **Bicuspid valves** – are found between the left atrium and left ventricle to prevent the back flow of blood from left ventricle to left atrium.
- ✓ **Semi lunar valves** – found at the entrance of pulmonary artery to prevent the back flow of the blood into the right ventricle.

Note: Blood always flaws from atrium to ventricles.

Heart beat and blood pressure

Contraction of the heart muscles is stimulated by specialized tissues called **pacemaker**. They are located on the right side of the heart where the contraction spreads to other parts of the heart.

The contraction of auricles and ventricles is called **systole**. This helps to force blood out of the heart.

The relaxation of the chambers of the heart (auricles and ventricles) is called **diastole**. This fills the chambers of the heart with the blood.

An instrument that is used to listen to the heart beat is called **stethoscope**. The average heart beat rate of an adult human is 72 times per minute. However, the beating of the heart can be affected by factors such as exercise, age, anger, drugs, etc.

During contraction of the ventricles, tricuspid and bicuspid valves close to prevent the back flow of blood in to the atria.

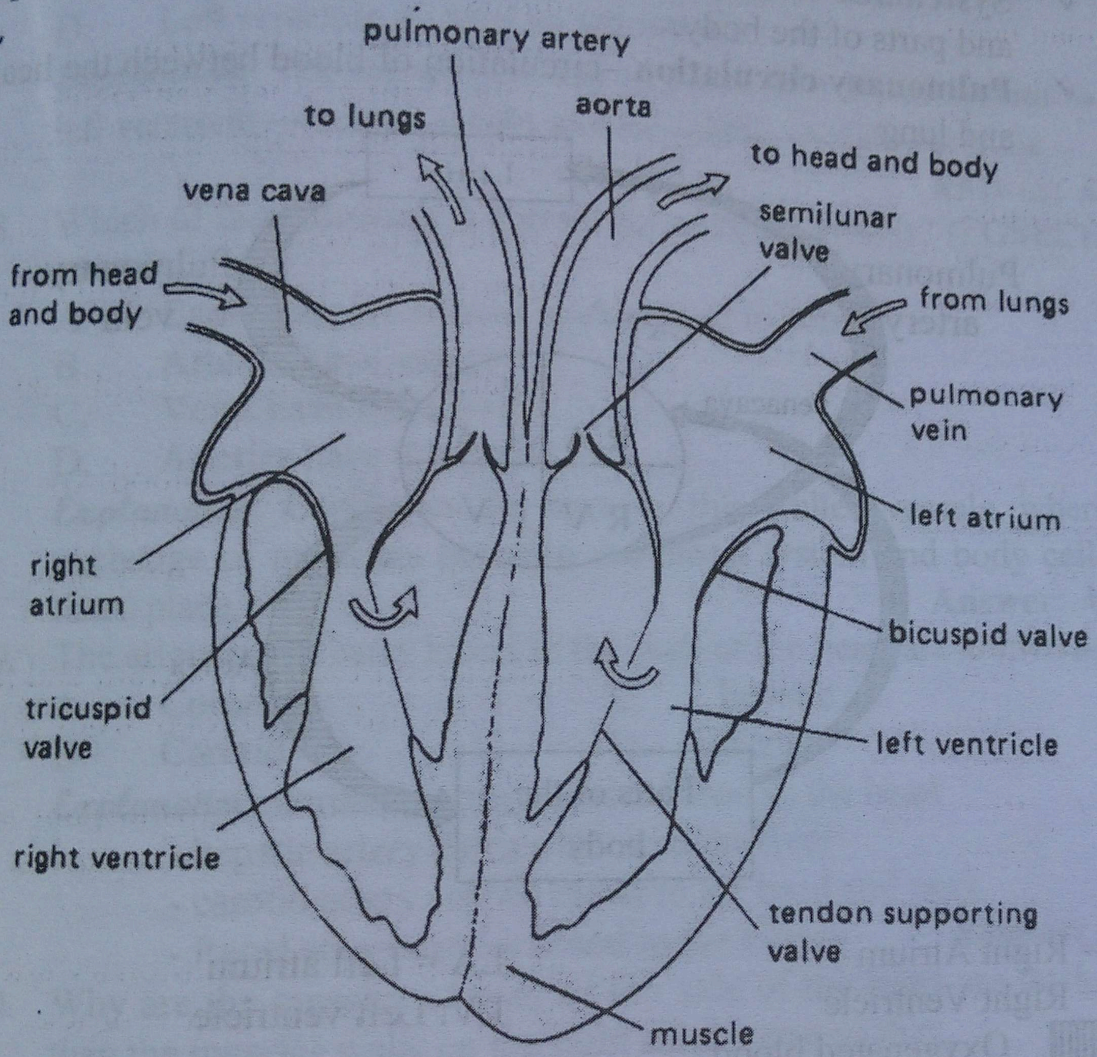


Fig. 3:5 structure of human heart

At normal condition human heart beats about 72 times/minute.

Blood pressure

Blood pressure a pressure that is exerted as a result of the contraction and relaxation of the muscles of the heart. These are systolic and diastolic pressure.

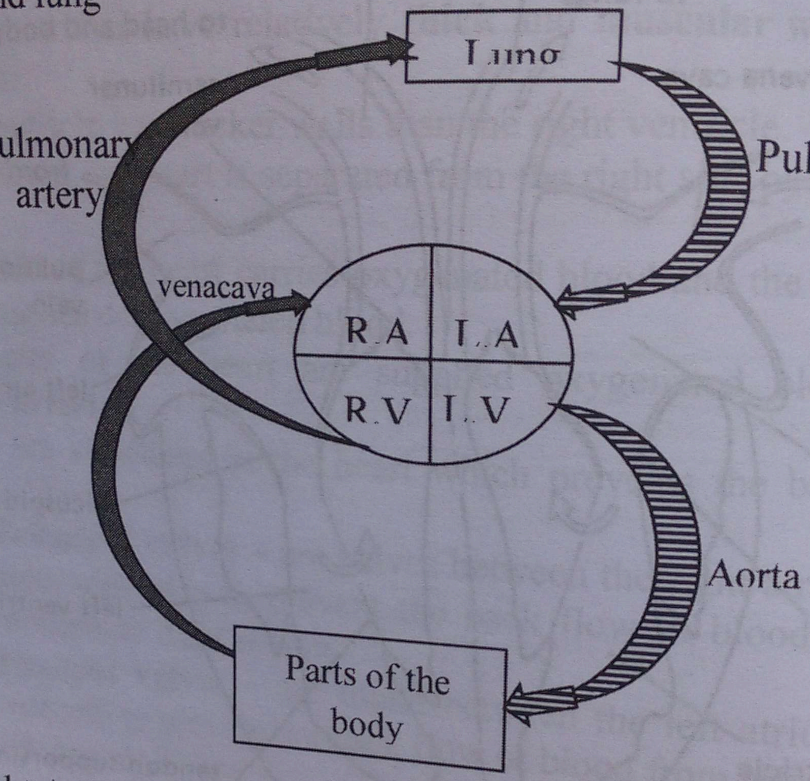
- ✓ **Systolic pressure** – is a pressure which is formed by the contraction of the muscles of the heart.
- ✓ **Diastolic pressure** – a pressure exerted by the relaxation of the muscles of the heart.

The normal blood pressure for human being is 120mmHg/80mmHg The nominator is systolic and the denominator is diastolic pressure.

Double circulation

Human circulatory system is called a **double circulation**. It involves two circulations:

- ✓ **Systematic circulation** – circulation of blood between the heart and parts of the body.
- ✓ **Pulmonary circulation** –circulation of blood between the heart and lung



Key

RA = Right Atrium
RV = Right Ventricle
LA = Left atrium
LV. Left ventricle
Oxygenated blood
Deoxygenated blood

Fig. 3:6 Double circulation

Pulse – is a wave of expansions and contraction of the artery walls that proceeds from the aorta along the arteries and arterioles. It is detected in parts of the body such as wrist, back of knees, ankle and neck where arteries are close to the skin surface.

Sphygmomanometer is an instrument that is used to measure blood pressure.

Stethoscope – an instrument that is used to listen to the heart beat

Illustrative Example

117. Which of the following pathways for the circulation of blood in human is correct?
   A. Left ventricle → body → right auricle
   B. Right auricle → lungs → left auricle
   C. Right auricle → body → left ventricle
   D. Left ventricle → lung → left auricle

Explanation: The correct pathway of blood flow is: left atrium → left ventricle → body → right atrium →lung → right ventricle

Answer: A

118. Which of the following is correct about blood vessels? (EGSECE 2004 E.C)
   A. Capillaries are sites of exchange of materials
   B. Arteries have valves
   C. Veins have thick walls
   D. Arteries have thin walls

Explanation: Capillaries are narrow, thin-walled vessels, where exchange of materials between the blood system and body cells takes place.

Answer: A

119. The arteries that carry blood to the wall of the heart are known as
   A. Coronary
   B. Carotid
   C. hepatic
   D. Renal

Explanation: coronary artery carries blood to the heart
- hepatic artery carries blood to the liver
- carotid artery carries blood to the head and neck
- Renal artery carries blood to the kidney

Answer: A

120. Why are the muscles of wall of left side of the heart are thicker than the muscles walls on the right side? Because they,
   A. Pump blood to relatively shorter distance
   B. Protect the heart from external damage
   C. Pump blood to relatively longer distance
   D. Receive much of the digested food to build them

Explanation: The right side of the heart pumps blood only to the lung, however the left part of the heart pumps blood to the whole parts of the body, thus it has thicker wall.

Answer: C

121. The transfer of oxygen to the body cells takes place in the
   A. Arterioles B. Artery C. capillaries D. vanules

Explanation: capillaries are thin walled blood vessels wh... exchange of materials between the body tissues and blood ta... place.

Answer:

122. Which one of the following series represents the correct pat... blood circulation?

A. Right atrium, right ventricle, lung, left atrium, le... ventricle, body

B. Left atrium, left ventricle, lung right atrium, righ... ventricle, body

C. Right atrium, lung, right ventricle, left atrium, body, le... ventricle

D. Left atrium, left ventricle, right atrium, right ventricle, lungs, body

Answer: A

For more Explanation look item number 92

123. Oxygenated blood leaves the human heart through the (EGSECE 1996 E.C)

A. Pulmonary vein
B. Pulmonary artery
C. venacava
D. aorta

Explanation: Oxygenated blood leaves the left part of the heart through **aorta** and deoxygenated blood leaves the right side of the heart through **pulmonary artery**.

Answer: D

124. Which two chambers of the human heart contain oxygenated blood

A. Right auricle and left auricle
B. Left auricle and right ventricle
C. Left auricle and left ventricle
D. Right auricle and right ventricle

Explanation: The left part of the heart carries oxygenated blood, while the right side of the heart carries deoxygenated blood.

Answer: C

Item number 125 based on the list of the four heart chambers given below. (EGSECE 2003 E.C)

1. Left auricle
2. Right ventricle
3. Left ventricle
4. Right auricle

125. Suppose you are ingested a chemical substance with your meal which specifically degrades the function of cardiac tissue. In which of the following orders does the degradation takes place?

A. 2,3,4,1
B. 4,2,1,3
C. 1,3,4,2
D. 4,1,3,2

Explanation: The flow of blood always takes place from auricle to ventricles. Thus the correct path way is:
Right auricle → right ventricle → left auricle → left ventricle

Answer: B

126. If the pulse rate of an adult man at rest is 18 times in 15 seconds what is the pulse rate of this man per minute?
A. 72 B. 108 C. 90 D. 54

Answer: A

Explanation: The heart beat rate of human being at rest is around 72 times/minute

127. Arteries have thicker and more elastic wall the veins. This is due to the fact that
A. Veins receive blood under pressure
B. Veins carry deoxygenated blood
C. Arteries receive blood under pressure
D. Arteries carry more digested end products

Explanation: Arteries receive blood which is pumped out of the heart as a result of systolic pressure. Systolic pressure has high pressure than diastolic pressure. **Answer: D**

Item number 128 based on the following structure involved in pulmonary circulation (EGSECE 2001 E.C)
1. Right atrium 4. Pulmonary artery
2. Pulmonary vein 5. Lung
3. Left ventricle 6. Left atrium 7. Right ventricle

128. The order of blood flow is
A. 1,7,2,5,4,6,3 C. 1,6,4,5,7,3,2
B. 1,7,4,5,2,6,3 D. 6,3,4,5,2,1,7

Answer: B

129. The chamber of the heart which pumps blood to all parts of the body (EGSECE 1999 E.C)
A. Right auricle C. left auricle
B. Right ventricle D. left ventricle

Explanation: The left ventricle pumps oxygenated blood to all parts of the body, while the right ventricle pumps deoxygenated blood to the lung **Answer: D**

130. Blood pressure is the pressure exerted by the blood on the wall of
A. Arteries C. ventricle
B. Auricles D. vein

Explanation: The blood pressure is created mainly as a result contraction and relaxation of left ventricle.

Answer:

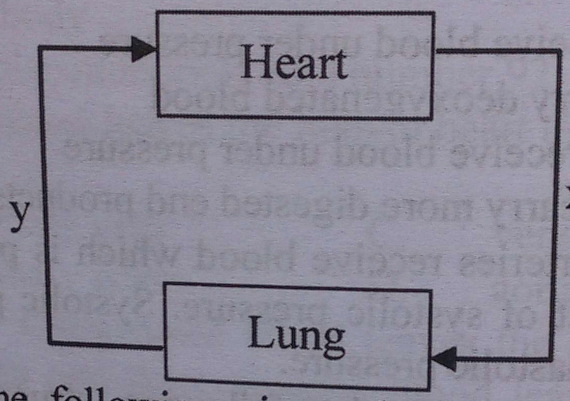
131. If one injects a blue dye into left ventricle, in which of following parts of the body will dye appear next?

A. Right auricle
B. Left auricle
C. pulmonary artery
D. right auricle

Explanation: According to the pathway of blood through blood vessels and the heart. Right auricle is nearest to right auricle than those which are given in the alternatives.

Answer:

Item number 132 is based on the following diagram about blood flow given below.



132. Which of the following given alternative is correct about the diagram?

A. Path 'y' carries oxygenated blood while path x carries deoxygenated blood
B. The flow starts from the lungs and ends in the heart
C. Path "x" carries oxygenated blood while path "y" carries deoxygenated blood
D. The diagram shows systematic blood circulation

Explanation: Blood that flows from lung to heart is oxygenated blood while blood that is carried from heart to lung is deoxygenated.

Answer: A

133. The structure that prevents the backflow of blood from right ventricle to the right atrium is known as

A. Tricuspid valve
B. Semi lunar valve
C. bicuspid valve
D. septum

Explanation: The back flow of blood from right ventricle to right auricle is prevented by **tricuspid valve,** while the back flow of blood from left ventricle to left auricle is prevented by **bicuspid valve.**

Answer: A

134. The movement of blood from the heart to the lung and back to the heart is known as

A. Pulmonary circulation
B. Double circulation
C. Closed circulation
D. open circulation

Explanation: **Pulmonary circulation** –is the flow of the blood between heart and lung.

Heart ⟸⟹ lung

Systematic circulation – is the flow of the blood between heart and parts of the body,

Heart ⟸⟹ parts of the body

Answer: A

135. Clearly differenciate between arteries and veins

Explanation:

Arteries	Veins
✓ Carry blood away from the heart	✓ Return blood back to the heart.
✓ Mostly carry oxygenated blood.	✓ Mostly carry deoxygenate blood.
✓ Have thick and elastic wall	✓ Have thin and less elastic wall
✓ Have no valves	✓ Have valves

136. What makes pulmonary artery different from other arteries.

Explanation: Most arteries carry oxygenated blood to different organs of the body. However, the **pulmonary artery** carries deoxygenated blood from the right ventricle to the lung.

137. Write the names of blood vessels that carry blood to the kidney, head and gut respectively.

Explanation:

✓ **Renal artery** carries oxygenated blood to the kidney.
✓ **Carotid artery** supplies oxygenated blood to the head and neck.
✓ **Mesenteric artery** supplies oxygenated blood to the gut.

138. Clearly discuss the role of the following parts of the human heart.

A. Pericardium
B. Septum
C. Pace maker

Explanation:

✓ **Pericardium** is the membrane that surrounds the heart. Thus, it provides protection.

- ✓ **Septum** prevents the mixing of oxygenated deoxygenated blood
- ✓ **Pacemaker** stimulates the beating of the heart.

139. What is blood pressure?

Explanation:

Blood pressure is the pressure that is exerted by the muscle contraction (systolic pressure) and relation (diastolic pressure) the heart.

140. Discuss the importance of:
    - ➢ Tricuspid valve
    - ➢ Bicuspid valve
    - ➢ Semi lunar valves

Explanation:

- ✓ **Tricuspid valves** – prevents the back flow of the blood from the right ventricle to right atrium.
- ✓ **Bicuspid valve** – prevents the back flow of the blood from the left ventricle to the right atrium
- ✓ **Semi lunar valves** prevent the back flow of the blood from the pulmonary artery and aorta back in to the heart.

141. What is the difference between systolic pressure and diastolic pressure

Explanation:

Systolic pressure is the pressure which is exerted as a result of muscular contracts of the heart chambers.

Diastolic pressure is the pressure which is exerted as the result of muscular relaxation of the heart chambers.

142. What is sphygmomanometer?

Explanation:

Sphygmomanometer is an instrument which is used to measure the blood pressure of an individual.

Post Test 5

Part I: Choose the best answer form the given alternatives.

1. The chamber of the heart that pumps blood relatively to a longer distance is:
    - A. Left atrium
    - B. Left ventricle
    - C. Right atrium
    - D. Left ventricle
2. The muscles of the heart are called:
    - A. Smooth
    - B. Cardiac
    - C. Skeletal
    - D. Elastic

3. Which of the following chambers of the heart carries oxygenated blood?
   A. Left and right ventricle
   B. Left atrium and left ventricle
   C. Right atrium and right ventricle
   D. Left and right atrium
4. The right side of the heart is separated from the left side by:
   A. Pericardium C. Septum
   B. Chamber D. Valve
5. The beating of the heart is detected by an instrument called:
   A. Sphygmomanometer C. Pacemaker
   B. Stethoscope D. Cardio scope
6. The valve that is found in between the right atrium and right ventricle is:
   A. Semi lunar valve C. Tricuspid valve
   B. Bicuspid valve D. Heart valve
7. Systematic circulation in humans is the flow of blood from :
   A. Heart $\rightleftharpoons$ Parts of the body
   B. Parts of the body $\rightleftharpoons$ Heart
   C. Heart $\rightleftharpoons$ Lung
   D. Lung $\rightleftharpoons$ Heart
8. The pressure of the blood is measured by an instrument called:
   A. Sphygmomanometer
   B. Stethoscope
   C. Thermometer
   D. Photo light
9. Which one of the following is untrue about the arteries?
   A. They are highly elastic
   B. They only carry oxygenated blood
   C. They always carry blood away from the heart
   D. They have thickest wall
10. All veins carry deoxygenated blood except:
   A. Hepatic vein
   B. Renal vein
   C. Pulmonary vein
   D. Subcalivian vein

11. During double circulation in humans, blood moves from the h... to lung for:
    A. Getting deoxygenated
    B. Getting oxygenated
    C. Removing oxygen
    D. Picking carbon dioxide

Part II: Match the following blood vessels with their functions.

Column A	Column B
1. Renal artery	A. Supplies oxygenated blood to the head and neck
2. Carotid artery	B. Carries deoxygenated blood back to the heart
3. Hepatic artery	C. Supplies oxygenated blood to the liver
4. Mesenteric artery	D. Carries oxygenated blood to the kidney
5. Vanacava	E. supplies oxygenated blood to the gut

The blood

Blood is a fluid tissue that is used to carry nutrients, respiratory gases, metabolic wastes and other substances. The amount of blood which is found inhuman body is 5.5 liters in average. It is made up of solid (45%) and liquid (55%) called **plasma**.

- ✓ The solid portion of the blood is made up of **red blood cells, white blood cells** and **platelets**.
- ✓ The liquid portion **plasma** is composed of largely water (90%) and dissolved substance (10%) like amino acids and glucose.

Red blood cells (Erythrocytes)

RBCs are disc shaped and non-nucleated cells which are used to transport **oxygen** and return **carbondioxide** back into the lung.
RBCS Consists of a red pigmented substance called **hemoglobin**.

- Hemoglobin carries oxygen in the blood
  Hemoglobin + oxygen ⟶ oxyhemoglobin
- Hemoglobin is a protein which also contains iron.

White bloo

White blood
are used to de
✓ Phago
micr
✓ Lymp
anti

Platelets (T
Platelets ar
initiating th

Human b

Human bl
major bloo

Antigens
blood ce
Antibod
the bloo

There a

Antibo
recogn
foreig
Agglu
group

White blood cell (leucocytes)

White blood cells are colorless, irregularly shaped and nucleated. They are used to defend the body against disease. WBCs can be classified as:

- ✓ **Phagocytes** – groups of WBC's that engulf or ingest infectious microorganisms
- ✓ **Lymphocytes** – group of WBC's which help to produce antibodies.

Antigens – are proteins that are foreign to the body

Platelets (Thrombocytes)

Platelets are colorless and non-nucleated. They play important role in initiating the blood clotting when blood vessels are cut or damage.

Human blood groups and transfusion

Human blood grouping is ABO blood grouping system. There are four major blood groups in humans. These are:

i. Blood group – A
ii. Blood group – B
iii. Blood group – AB
iv. Blood group O

Antigens – are special proteins which are found on the surface of red blood cells.

Antibody – are proteins which are produced and found in the plasma of the blood.

- ➢ They can recognize foreign antigens and destroy foreign cells (antigens).

There are two types of antigens. Namely; antigen – A and antigen – B

- ✓ **Antibodies** – are proteins which are produced by the body to fight against antigens. There are also two types of antibodies. These are antibody – a and antibody – b

Antibody –a recognizes antigen – A as foreign protein and antibody – b recognizes antigen – B as foreign protein. This results in destruction of foreign cells.

Agglutination – is incompatibility of antigen and antibody in blood groups. This results in clumping and sticking together of blood cells.

Table 3.6: Antigens and antibodies of different blood groups and transfusion

Blood group	Antigens of red blood cell	Antibodies in the plasma	Donated to	Receive from
A	A	B	A and AB	A and
B	B	A	B and AB	B and
AB	A and B	None	AB only	All gro
O	None	A and B	All groups	O only

Thus, blood group 'O' is called **universal donor** and blood group 'A is called **universal recipient.**

Problems related to circulatory system

Anemia – Is caused when there are few red blood cells or if the le of hemoglobin is less.
- It's most commonly caused due to lack of iron in the die So, it's treated by eating iron rich diets like liver, mea eggs and green leafy vegetables.

Hypertension – high blood pressure. In which the systolic an diastolic pressure increases.
- The high blood pressure is when it's above 140/90mmHg.
- It can be caused by various reasons such as age over weightiness, high consumption of salt, drug addiction, stress, kidney problem, diabetes etc.
- It can be reduced and controlled by:
  - Eating less fat and salt
  - Regular exercise
  - Avoid use of drugs
  - Living easy life or changing life style regular checkups
  - Using diuretics which are prescribed by a doctor or health professional

Illustrative Examples

143. Which of the following types of blood do not have antigens in its red blood cells?
A. O B. A C. B D. AB

Explanation:
✓ Blood group 'O' have no antigens in its red blood cells.
✓ Blood group 'AB' has antigens A and B on the surface of its red blood cells

✓ Blood group 'A' has antigen A in its red blood cells.
✓ Blood group 'B' has antigen B in its red blood cells.

Answer: A

144. Which of the following blood transfusion is compatible?

	Donor	Recipient
A.	A	B
B.	B	A
C.	O	B
D.	AB	O

Explanation: Blood group 'O' can be donated to all groups. The correct compatibility of the blood is

Donor	Recipient
A	A or AB
B	B or AB
AB	AB only
O	All

Answer: C

145. Blood transfusion between blood type A and blood type B caused agglutination. This is because (EGSECE 2002 E.C)
A. Type A has antigen B while type B has antigen A.
B. Both types of bloods have antigen B.
C. Both types of blood have antibody A.
D. Type A has antigen A while type B has antibody A.

Explanation: The fusion of antigen –A and antibody a causes agglutination.

Answer: D

146. Which of the following is correct about human blood cells? (EGSECE 1197)
A. White blood cells contain hemoglobin.
B. Red blood cells are fewer than white blood cells
C. Red blood cells defend the body against disease causing organism
D. Mature red blood cells do not passes nuclei

Explanation:
It is red blood cells that contain hemoglobin.
✓ Red blood cells are smaller and numerous than white blood cells.
✓ White blood cells defined the body against disease
✓ Mature RBC's have no nuclei.

Answer: D

147. From which blood group can blood be transferred regardless the recipient blood group?
A. AB B. O C. A D. B
Explanation: Blood group 'O' can be donated to all blood groups. Therefore, it's called **universal donor.** **Answer:**

148. Suppose a person lost a lot of blood due to a car accident. If his blood group is "O", for a successful blood transfusion the donor blood should be
A. A B. O C. B D. AB
Explanation: A person with blood group 'O' can receive blood only from blood group 'O'. **Answer: B**

149. A disease of blood which is inherited from parents is
A. Leukemia C. varicose vein
B. Hemophilia D. elephantiasis
Explanation: Hemophilia is a disease of blood which is characterized by un ability of the blood to clot after cut or damage on the skin which may result in excessive loss of blood.
- Hemophilia is hereditary disease that can be inherited from parents to their children. **Answer: B**

150. Abebe's Father used to suffer from blood clotting problem, Abebe also suffers from the same case. What is the most probable cause of their problem?
A. Elephantiasis C. Hemophilia
B. Varicose –vein D. leukemia
Explanation: Hemophilia is a blood clotting problem that can be inherited from parents to their children. **Answer: C**

151. Which of the following human blood type lacks antibody? (EGCSE 2001 E.C)
A. O B. B C. A D. AB
Explanation: Blood type AB has no antibody in its plasma, thus it can receive blood from all blood types **Answer: D**

152. Suppose the blood pressure of a person is found to be 140/95 which of the following is true?
A. The person has a problem of high blood pressure
B. The person needs to increase his/her weight
C. The denominator is the pressure resulting from contraction of left ventricle.
D. The nominator indicates diastolic pressure

Explanati
walls of t
heart. Th
140/95
hyperten
Item number 1
type, antibody
2004)

Blood type	
A	
B	
AB	
O	

153. "x" rep
A.
B.
Expla
blood
154. Which
clottin
A.
B.
Expl
prote
cut o
155. A di
bloo
A.
B.
Exp
whi
can
156. Wh
A.
B.
C.
D.

Explanation: Systolic pressure is created by contraction of the walls of the heart and diastolic pressure is relaxation of walls the heart. The normal blood pressure of human is 120/80. Thus 140/95 indicates high blood pressure which may lead to **hypertension.**

Answer: A

Item number 153 is based on the following table which shows blood type, antibody and antigen on the surface of red blood cells (EGSECE 2004)

Blood type	Antibody in plasma	Antigen on the surface of RBC
A	B	A
B	A	B
AB	No antibody	A and B
O	A and B	X

153. "x" represents
   A. Antigen B
   B. No antigen
   C. antigen A
   D. Antigen A and B

Explanation: Blood type 'O' has no antigen on its surface of red blood cells.

Answer: B

154. Which of the following blood cells are involved in blood clotting? (EGSES 1993 E.C)
   A. Red blood cell
   B. Platelets
   C. white blood cells
   D. lymph

Explanation: Platelets are cells that are responsible for producing proteins which are used to clot the blood after we get injured or cut on our skin.

Answer: B

155. A disease condition caused by uncontrolled production of white blood cells is.
   A. Hemophilia
   B. Elephantiasis
   C. leukemia
   D. anemia

Explanation: leukemia is uncontrolled production of WBCs which causes a decline in number of RBS. Its also called blood cancer.

Answer: A

156. Which of the following is true about the blood vessels?
   A. Pulmonary veins carry deoxygenated blood.
   B. Pulmonary arteries carry deoxygenated blood.
   C. Veins carry blood away from the heart.
   D. Arteries carry blood towards the heart.

Explanation:

- Pulmonary veins – carry oxygenated blood from lu... heart (left auricle)
- Pulmonary arteries – carry deoxygenated blood fro... heart (right ventricle) to the lung.
- Arteries are blood vessels that carry blood away fro... heart
- Veins carry blood towards the heart.

157. Which of the following are the waste products of anaero... respiration by yeast cells? **Answer:**

A. Ethanol and carbondioxide
B. Water and lactic acid
C. Lactic acid and ethanol
D. Carbon dioxide and water

Explanation:

- Anaerobic respiration – is a kind of respiration that do... require oxygen.
  1) Anaerobic respiration in yeast

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \xrightarrow{\text{yeast}} \underset{\text{ethanol}}{2C_2H_5OH} + \underset{\text{carbondioxide}}{2CO_2} + 2ATP$$

  2) Anaerobic respiration in animals.

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \longrightarrow \underset{\text{lactic acid}}{2C_3H_6O_3} + 2ATP$$

158. A grade 9 student conducted an experiment in the biology laboratory. First she put a sample of glucose powder and water in to a test tube. Then she added a few drops of Benedict's solution in to the test tube and placed it in boiling water. What was the most probable color she observed within the test tube? **Answer: A**

A. Blue B. purple C. Red D. Orange

159. Which of the following statements is correct about simple, double, and complex sugars? **Answer: D**

A. Sucrose is simple sugar, fructose is double sugar, and starch is complex sugar.
B. Complex sugars are non sweet and do not dissolve in water while simple and double sugars are sweet and dissolve in water.

C. Simple sugars are sweet and dissolve in water while double and complex sugars are non sweet and do not dissolve in water.

D. Simple and double sugars are used as storage molecules while complex sugars are structural molecules.

Explanation: A carbohydrate can be classified in to three based on the number of sugar units they contain.

Simple sugar	Double sugar	Complex sugar

• Contain one sugar unit
• Have sweet taste
• Soluble in water
• Examples are
  • Glucose
  • Galactose
  • Fructose
	• Consist of two sugar units
	• Have sweet taste
	• Soluble in water
	• Example are:
	  • Sucrose
	  • Lactose
	  • Maltose
		• Consists of many sugar units.
		• Have no sweet tast
		• Are insoluble in water
		• Used as storage form of carbohydrate in organisms
		• Examples:
		• Starch
		• Glycogen
		• Cellulose
		• chitin

Answer: B

160. Which of the following comparisons about aerobic and anaerobic respiration is correct?

A. The waste products of anaerobic respiration are CO_2 and H_2O while that of aerobic respiration is lactic acid.

B. Energy produced by aerobic respiration in much higher than that produced by anaerobic respiration.

C. Aerobic respiration occurs in the cytoplasm while anaerobic respiration occurs in the mitochondria.

D. Aerobic respiration occurs in animals cells while anaerobic respiration occurs in plant cells.

Explanation: There are two types of respiration:

- Aerobic respiration
- Anaerobic respiration

Aerobic respiration	Anaerobic respiration
➢ Requires oxygen ➢ Produces much more energy, (36/3ATPs from 1 glucose) molecule. ➢ Occurs in mitochondria	➢ Does not require oxygen ➢ Produces less amount of energy. (2ATPs from glucose molecule). ➢ Occurs in cytoplasm of cell

Answer:

161. The reaction that is the immediate source of energy for cellular activities is:

A. $ADP + P + Energy \rightarrow ATP + H_2O$

B. $Glucose + oxygen \rightarrow CO_2 + H_2O + Energy$

C. $Glucose \rightarrow lactic\ acid + energy$

D. $ATP + H_2O \rightarrow ADP + p + Energy$

Explanation:

- ATP (adenosine triphosphate) is a molecule which is composed of ADP (Adenosine diphosphate) phosphate and energy that is released from break down of glucose molecule
- ATP – serves as the immediate (cash) source of energy for the cell.

$ADP + P + Energy \longrightarrow ATP$

Answer: A

162. Which of the following components of the blood play important role in clotting blood during injury?

A. Platelets
B. Red blood cells
C. White blood cells
D. plasma

Explanation:

- **Platelets** – are cells which are used to clot blood during cut or injury in order to prevent excessive loss of blood from the body.
- **Red blood cells** – Transportation of nutrients, oxygen and other use full substances in the body.
- **White blood cells** – protection of body against disease.
- **Plasma** – the liquid portion of the blood.

Answer: A

163. Why is the double circulation so important for warm blooded animals? Because it
A. Increases pumping efficiency of the heart
B. Supplies fully oxygenated blood to all body cells.
C. Supplies all cells with sufficient amount of nutrients.
D. Carries blood directly from the lungs to the body cells.

Explanation:
- **Double circulation** – involves two major circulations. These are;
  - **Pulmonary circulation** – the flow of blood from heart to lung and again back to the heart.
  - **Systematic circulation** – Is a flow of blood from heart to different parts of the body and back to the heart.
- Therefore, double circulation is very efficient to supply fully oxygenated blood to our cells.

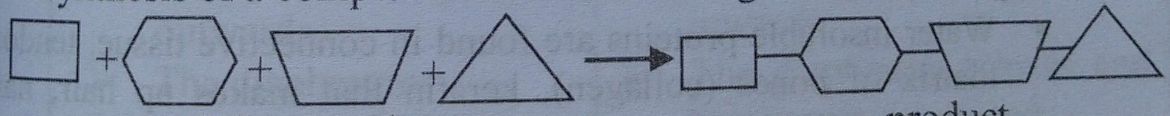
Answer: B

164. What are antigens? They are proteins which
A. Allow red blood cells to accommodate more oxygen.
B. Destroy foreign cells to the body.
C. Play regulatory role in cells.
D. Recognize cells from different organisms.

Explanation: Antigens are proteins which are found on the surface of all cells. They allow cells to regencies each other and also recognize cells from different organisms.

Answer: D

165. This item is based on the following diagram which represents the synthesis of a complex molecule in an organisms



Building blocks product

Which of the following option represents the building blocks and the production?
A. Sugar molecules → ATP
B. DNA molecules → part of starch
C. Starch molecules → Glucose
D. Amino acid molecule → part of protein

Explanation:
- ATP molecule is built from ADP, phosphate and energy from break down of glucose.

- Starch is a polysaccharide (complex sugar) which composed of a long chain of sugar (glucose) molecules.
- Protein is organic molecule which is made up of building blocks called **amino acids**

Answer:

166. Which of the following is the role of hydrochloric acid secre by the wall of stomach?
   A. Facilitates the digestion of lipids in the stomach
   B. Breaks down carbohydrates in to smaller sugar molecule
   C. Controls the action of sphincter muscle at the entrance stomach.
   D. Creates conducive environment for the action of pepsin

Explanation:

- Protease enzymes (pepsin and rennin) which are found in the stomach work suitability at acidic environment. This describes the property of enzymes which says that;
  - Enzymes are sensitive to PH.
- Therefore HCI (hydrochloric acid) makes the stomach suitability acidic for action of enzymes (Renin and pepsin). It is also used to kill germs.

Answer: D

167. This item is based on the following list of protein substances.
   1) Antitoxins
   2) Matrix of bone
   3) Silk of spider web
   4) Enzymes
   5) Hormones
   6) Keratin

Substances that are made up of water soluble proteins include:
   A. 2,3,4
   B. 1,5,6
   C. 1,4,5
   D. 2,5,6

Explanation:

- Water insoluble proteins are found in connective tissue, tendons, matrix of bones (collagen), keratin that makes up hair, nails, horns and Feathers.
- Water soluble proteins form antibodies, enzymes, some hormones and maintaining structure of cytoplasm.

168. This item is based on the following table which shows the body weight of four individuals.

Answer: C

No.	Name	Body weight (kg)	Height (m)
1	Abebe	80	1.4
2	Chaltu	50	1.7
3	Azeb	65	1.6
4	Mohammed	90	1.5

Which of the individuals seems to be healthy depending on his/her body weight and height?

A. Azeb
B. Chaltu
C. Mohammed
D. Abebe

Explanation:

- Body /mass index (MI) = $\frac{weight}{(height)^2}$
- The average body/ mass index of most people in average is 20 – 30. However, the BMI below 18.5 or above 35 may be signs of some health problems.

Accordingly;

BMI of Abebe = $\frac{80}{(1.4)} = 40.8$ BMI of Azeb = $\frac{65}{(1.5)^2} = 25.39$

BMI of chaltu = $\frac{50}{(1.7)^2} = 17.3$ BMI of Mohammed = $\frac{90}{(1.5)^2} = 40$

Answer: A

169. Which of the following events is true during inhalation?

A. The pressure inside the chest is lower than outside
B. The ribs move down and in
C. The diaphragm muscles relax.
D. External intercostals muscle relax.

Explanation:

During inhalation	During exhalation
➢ The ribs move up and outwards. ➢ The diaphragm contracts and becomes flatten. ➢ Intercostals muscle contracts ➢ The pressures inside the chest is lower ➢ The lung inflates (fills with air)	➢ The ribs move down and inwards ➢ The diaphragm relaxes and become dome shaped ➢ Intercostals muscle relax ➢ The pressure becomes higher ➢ The lung deflates (shrinks)

Answer: A

170. Which of the following carbohydrates in mainly found in sugar?

A. Sucrose
B. Lactose
C. Glucose
D. Maltose

Explanation:

- Sucrose, Maltose and lactose are dauble sugars.
- Glucose is a simple sugar.
- Double sugars are composed of two sugar units. Thus,

Glucose + Galactose ⟶ lactose (milk sugar)
Glucose + Fructose ⟶ sucrose (cane sugar)
Glucose + glucose ⟶ Maltose (Sugar found in germinating seeds like barley)

Answer:

171. Which of the following is correct about the effect of altitude on breathing rate? At higher altitude breathing rate:

A. Decreases because of the atmospheric pressure.
B. Increases because of the low level of oxygen in the air.
C. Decreases because of high level of oxygen in the air.
D. Increases because of the high level of carbondioxide in the air.

Explanation:

- As altitude increases the air becomes less dense. This means, the amount of air in a given volume decreases. Therefore, in order to get more and more oxygen, a person needs to breathe faster and deeper at a place of higher altitude.

116. Differentiate between: **Answer: D**
Systematic circulation and pulmonary circulation

Explanation:

✓ **Systematic circulation** is the flow of blood from left auricle to different parts of the body and again
✓ **Pulmonary circulation** is the flow of blood from right ventricle to the lung and again back to the heart.

117. Clearly discuss the role of the following blood cells in the body
✓ Red blood cells
✓ White blood cells
✓ Platelets

Explanation:

- ✓ **Red blood cells** are used to transport substances like oxygen, nutrients, hormones, etc in the body.
- ✓ **White blood cells** provide protection against disease.
- ✓ **Platelets** – play role in blood clotting when our blood vessels are – cut.

118. What is agglutination?

Explanation:

Agglutination: is clumping together of blood cells which is caused due to the in compatibility of blood types.

- ✓ It results in death. Thus, checking the donors and recipients blood group is very important in order to avoid agglutination.

119. Which of the following food classes is correctly matched with its source of food? (EGSECE, 2006 E.C)

A. Lipid ⟶ milk, butter oil
B. Mineral ⟶ carrot, cereal, yeast
C. Protein ⟶ Injera, fetira, honey
D. Vitamin ⟶ salt, green vegetable, bread.

Explanation: The correct match of nutrients and their source of food is:

Nutrient	Food source
Lipid	Butter, fatty meat, oil, ground nut, etc.
Mineral	Salt, liver, legumes, fish, milk, sea foods
Protein	Milk, egg, fish, chicken meat, etc
Vitamin	Carrot, diary product, cereals, fruits, etc

Answer: A

120. Red blood cells carry oxygen all over your body and supply it to your body cells. This is because red blood cells have: (EGSECE, 2006 E.C)

A. Hemoglobin
B. Antibodies
C. nucleus
D. antigens

Explanation:

Hemoglobin is a red pigmented substance that is used to carry oxygen in the blood. It is mainly composed of iron. Thus, the lack of iron in the diet causes **anemia.**

121. This item is based on the following steps of blood transport.

A. The right ventricle contracts and forces blood out of the heart in to the lung.

B. Oxygenated blood returns to the left side of the heart and the left atrium fills up.

C. The left ventricle contracts forcing oxygenated blood out of the heart to around the body.

D. The left atrium contracts farcing blood in to the left ventricle.

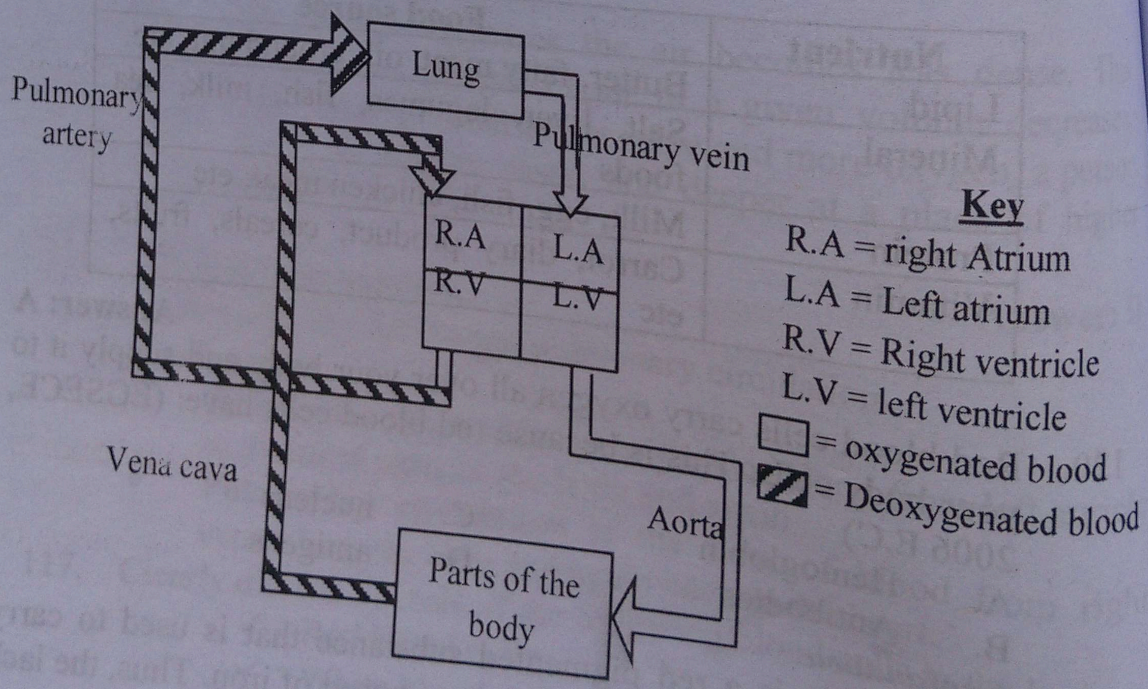
E. Deoxygenated blood comes into the right atrium of the heart from the veins to the body.

F. The right atrium contracts and forces blood in to the right ventricle.

Which of the following is the correct path of blood in the heat? (EGSECE, 2006 E.C)

A. E ⟶ C ⟶ A ⟶ D ⟶ F ⟶ B
B. E ⟶ A ⟶ C ⟶ D ⟶ B ⟶ F
C. E ⟶ A ⟶ B ⟶ F ⟶ C ⟶ D
D. E ⟶ F ⟶ A ⟶ B ⟶ D ⟶ C

Explanation: The correct path way of blood can be diagrammatically shown as:



Answer: D

122. Which of the following methods is used to prevent hypertension? (EGSECE, 2006 E.C)
A. Eating foods that have iron source.
B. Eating canned foods.
C. Doing regular exercise
D. Using vitamins.

Explanation:
- Hypertension is a non – infectious disease which is characterized by high blood pressure. Thus, it can be controlled by:
  - ✓ Eating less salt.
  - ✓ Reducing the level of cholesterol
  - ✓ Regular exercise
  - ✓ Living un stressful and healthy life
  - ✓ Avoid using drugs.

Answer: C

123. The three types of human blood vessels are: (EGSECE, 2006 E.C)
A. Aorta, vencava and capillaries
B. Artery, vein and capillaries.
C. Hepatic artery, pulmonary artery and renal artery.
D. Hepatic vein, pulmonary vein and renal vein.

Explanation:
In human circulatory system, there are three major types of blood vessels. These are:
- ➢ Arteries
- ➢ Veins
- ➢ Capillaries

Answer: B

124. Which of the following functions of blood vessels is true? (EGSECE, 2006 E.C)
A. Veins always carry blood towards the heart.
B. Arteries always carry oxygenated blood.
C. Pulmonary vein carries deoxygenated blood from lungs to the heart.
D. Umbilical vein carries deoxygenated blood from placenta to the fetus.

Explanation:

Veins	Arteries
• They carry blood towards the heart. • Have relatively thin and less elastic wall. • Mostly carry deoxygenated blood except pulmonary vein and umbilical vein	• They carry blood away from the heart • They have thick and elastic wall. • Mostly carry oxygenated blood except pulmonary artery and umbilical artery.

Answer: A

125. Assume that you have four test tubes A,B,C and D which contains equal amount of unknown food solutions. You added equal amount of ethanol and shook the tubes. Then you poured the contents in to other test tubes, a white cloudy layer formed only in test tube "C". What will be the food solution in test tube "C"? (EGSECE, 2006 E.C)

A. Vitamin C
B. Protein
C. Lipid
D. starch

Explanation:

In order to carry out a test for the presence of lipid in a given food sample, ethanol is used as a reagent which forms a milky or cloudy suspensions when lipid is added in to it.

126. Which of the following is true about balanced diet? (EGSECE, 2006 E.C)

Answer: C

A. Taking only very few foods is enough in order to get a balanced diet.
B. Getting enough food from major food classes is required in order to get a balanced diet.
C. Eating only fruits is sufficient in order to get a balanced diet.
D. Eating injera every day is enough in order to get a balanced diet.

Explanation:

A balanced diet involves getting the right type and precise amount of food from all the different groups of nutrients. This helps to keept body in a healthy and normal condition.

Answer: B

127. Which
mouth?
A.
B.
C.
D.
Expla
digesti

128. The
(EGS
A.
B.
Exp
In h

129. Wh
the
A.
B.
Ex

127. Which of the following digestion activities takes place in the mouth? (EGSECE, 2006 E.C)
A. Absorption of carbohydrates as glucose.
B. Production of hydrochloric acid by salivary glands.
C. Digestion of proteins to amino acids.
D. Breaking down of food in to smaller pieces.

Explanation: Mouth is a place where both physical (mechanical) digestion and chemical digestion takes place.

- The chemical digestion in the mouth can be indicated as:

$$\text{Starch} + H_2O \xrightarrow{\text{amylase}} \text{Maltose.}$$

- The physical digestion involves breaking down of food in to smaller pieces

Answer: D

128. The two human teeth types that are used for biting are: (EGSECE, 2006 E.C)
A. Canines and premolars
B. Incisors and premolars
C. Molars and premolars
D. incisors and canines

Explanation:
In humans there are four major types of teethes. These are:

- **Incisor** – are used for biting and cutting lamp of food
- **Canine** – are used for biting and tearing flesh.
- **Premolar** – are used for grinding and chewing food
- **Molar** – are used for grinding and chewing food.

Answer: D

129. Which of the following enzymes uses to digest carbohydrates in the mouth? (EGSECE, 2006 E.C)
A. Amylase
B. Lipase
C. Maltase
D. Trypsin

Explanation:

Enzyme	Reaction catalyzed	Organ
Amylase	$\text{Start} + H_2O \longrightarrow \text{Maltose}$	Mouth and duodenum
Lipase	$\text{Lipid} \longrightarrow$ fatty acid and glycerol	Duodenum and ileum
Maltase	$\text{Maltose} \longrightarrow$ glucose + galactose	Ileum
Trypsin	$\text{Peptide} \longrightarrow$ peptone	Duodenum

Answer: A